INDIAN INSTITUTE OF TECHNOLOGY MADRAS

MODERN APPLICATION DEVELOPMENT 1 (MAD 1 / APPDEV 1)

Comprehensive Past Year CBT Examination Question Bank (301 Authentic Questions)

Curriculum Week	Topic Description	Questions
Week 1	Internet Architecture, HTTP & Network Calculations	50 Qs
Week 2	HTML5, CSS3, Box Model & Form Controls	115 Qs
Week 3	Backend Python, Flask Essentials, Routing & Request Handling	78 Qs
Week 4	Jinja2 Templating, Dynamic Rendering, SQLite & CRUD Applications	58 Qs
Total	**Verified Authentic Past Exam Questions Across All 4 Weeks**	**301 Questions**

WEEK 1: INTERNET ARCHITECTURE, HTTP & NETWORK CALCULATIONS

- **1.1 Web & Internet Basics (Client-Server Architecture, Protocols & Servers) (10 Questions)**

MAD1_007 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

Consider the following code:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    p {
      color: blue;
    }
    #highlight {
      color: green;
    }
    .important {
      color: red;
    }
  </style>
</head>
<body>
  <p class=  'tant" id="highlight">This is a paragraph.</p>
</body>
</html>
```

What will be the color of the paragraph text when rendered in a browser?

Options :

(A) Blue

(B) Green

(C) Red

(D) The browser default (black)

Official Answer Key: 6406534366204

MAD1_017 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

THIS IS QUESTION PAPER FOR THE SUBJECT "MODERN APPLICATION DEVELOPMENT 1"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

(B) YES

(C) NO

Sub-Section Number : 2

(D) 64065348772

Question Shuffling Allowed : Yes

Question Number : 106 Question Id : 640653345241 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructior Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following statement is true about MVC architecture?

Options :

(E) The view redirects the incoming request to model.

Official Answer Key: 6406531148333

MAD1_028 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz 1 Jan23 .pdf

Match the component in column A with its correct feature given in column B.

Column A	Column B
1. Telephone Networks	a. Wires occupied only when data to be sent
2. Packet Switched Networks	b. First public packet switched network
3. ARPANET	c. Physical wires tied up for duration of call even if nothing said
4. TCP	d. Establishes reliable communications

Options :

(A) 1 - a, 2 - b, 3 - c, 4 - d

(B) 1 - b, 2 - d, 3 - a, 4 - c

(C) 1 - d, 2 - c, 3 - b, 4 - a

(D) 1 - c, 2 - a, 3 - b, 4 - d

Official Answer Key: 6406531562001

MAD1_035 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz 1 Jan23 .pdf

Match the component in column A with its correct feature given in column B.

Column A	Column B
1. Server	a. Data Pipeline
2. Client	b. Explicit Server(s), Explicit Client(s)
3. Client-Server Architecture	c. User Interaction
4. Network	d. Provides data on demand

Options :

(A) 1 - a, 2 - b, 3 - c, 4 - d

(B) 1 - b, 2 - d, 3 - a, 4 - c

(C) 1 - d, 2 - c, 3 - b, 4 - a

(D) 1 - c, 2 - a, 3 - b, 4 - d

Official Answer Key: 6406531561996

MAD1_093 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz1 May23 .pdf

What is the correct sequence of working of Client-Server Architecture.

a. The client asks for the IP address for the particular server from the Domain Name system.
b. To that IP address, the client sends the request to the particular server with that port number that is specified to the particular application and then server responds
c. The DNS server responds with the IP address.
d. The response message is received by the client and based on that port number, the response packet is consumed by the application to which it belongs.

Options :

(B) c → a → b → d

(C) a → b → c → d

(D) a → c → b → d

(E) None

Official Answer Key: 6406531929705

MAD1_100 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

Consider the following code:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    p {
      color: blue;
    }
    #highlight {
      color: green;
    }
    .important {
      color: red;
    }
  </style>
</head>
<body>
  <p class=  'tant" id="highlight">This is a paragraph.</p>
</body>
</html>
```

What will be the color of the paragraph text when rendered in a browser?

Options :

(A) Blue

(B) Green

(C) Red

(D) The browser default (black)

Official Answer Key: 6406534364662

MAD1_115 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

Consider the following code:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    p {
      color: blue;
    }
    #highlight {
      color: green;
    }
    .important {
      color: red;
    }
  </style>
</head>
<body>
  <p class=  tant" id="highlight">This is a paragraph.</p>
</body>
</html>
```

What will be the color of the paragraph text when rendered in a browser?

Options :

(A) Blue

(B) Green

(C) Red

(D) The browser default (black)

Official Answer Key: 6406534365433

MAD1_255 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf

Which of the following statements regarding dynamic webpages is/are true?

Options :

(E) The page is already present at the server even before a user requests it.

(F) The page generally maintains the same content every time the user requests

(7) The page is generated on demand by a program or a request from a browser.

(8) On every request, every new page created may be different from the last.

Official Answer Key: 6406532239849, 6406532239850

MAD1_257 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

Which of the following statements regarding dynamic webpages is/are true?

Options :

(E) The page is already present at the server even before a user requests it.

(F) The page generally maintains the same content every time the user requests

(7) The page is generated on demand by a program or a request from a browser.

(8) On every request, every new page created may be different from the last.

Official Answer Key: 6406532240945, 6406532240946

MAD1_259 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN4 EXAM QPD4 29 Oct.pdf

Which of the following statements regarding dynamic webpages is/are true?

Options :

(E) The page is already present at the server even before a user requests it.

(F) The page generally maintains the same content every time the user requests

(7) The page is generated on demand by a program or a request from a browser.

(8) On every request, every new page created may be different from the last.

Official Answer Key: 6406532241697, 6406532241698

• 1.2 Data Transmission, Bandwidth, RTT & Propagation Delay Calculations (17 Questions)

MAD1_010 | Type: SA | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf, IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

We want to use an online grammar checking tool, which takes each word from the client and sends it to the web server. The available network bandwidth between the client and the server is 2 Gbps. What should be the maximum allowed size(in MBs) of a document containing the word sent from the client to the server per second for the grammar check? (Use: 1KB= 1000Bytes, 1MB=1000KB and so on.)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

Question Number : 74 Question Id : 6406531298443 Question Type : SA
Correct Marks : 3

Official Answer Key: 250

MAD1_023 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

For a network bandwidth of 8 Gbps, what should be the size of each request if 5000 such requests are to be sent over the network per second? [**Use these relations:** 1 Byte = 8 bits, 1 KB = 1000 Bytes, 1 MB = 1000 KBs and so on.]

Options :

(A) 1.6 KB

(B) 200 KB

(C) 1.6 MB

(D) 200 MB

Official Answer Key: 6406531148388

MAD1_039 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024.pdf, IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024.pdf

A certain network connection with a bandwidth of 12 Mbps allows making 4000 requests to a server per second. If each request is modified to have an additional data of 6 Kb, what should be the additional bandwidth required to maintain the rate of 4000 requests per second?

Options :

(C) 12 Mbps

(D) 24 Mbps

(E) 36 Mbps

(F) 48 Mbps

Official Answer Key: 6406532467947

MAD1_056 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

A server S requests to retrieve data from two data centers DC_1 and DC_2, which are located at distances of D_1 and D_2 meters respectively from the server S. The speed of data travelling through the medium that connects S with DC_1 is m_1 m/sec and that connects S with DC_2 is m_2 m/sec. With the given media for data transfer between S $\rightarrow DC_1$ and S $\rightarrow DC_2$, it takes an additional 200 milliseconds for the data to reach S from DC_2 than it takes from DC_1 if the complete response is ready only when the data is available to S from both the datacenters. Figure out the value of m_2 (in m/sec) in terms of D_1, D_2 and m_1. [Note: The data centers send responses only when they receive a request from the server]

Options :

(C) $m_2 = \frac{D_2 m_1}{D_1 - 0.2m_1}$

(D) $m_2 = \frac{D_2 m_1}{D_1 + 0.2m_1}$

(E) $m_2 = \frac{2D_2 m_1}{2D_1 - 0.2m_1}$

(F) $m_2 = \frac{2D_2 m_1}{2D_1 + 0.2m_1}$

Official Answer Key: 6406533774187

MAD1_070 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

A server S requests to retrieve data from two data centers DC_1 and DC_2, which are located at distances of D_1 and D_2 meters respectively from the server S. The speed of data travelling through the medium that connects S with DC_1 is m_1 m/sec and that connects S with DC_2 is m_2 m/sec. With the given media for data transfer between S $\rightarrow DC_1$ and S $\rightarrow DC_2$, it takes an additional 200 milliseconds for the data to reach S from DC_2 than it takes from DC_1 if the complete response is ready only when the data is available to S from both the datacenters. Figure out the value of m_2 (in m/sec) in terms of D_1, D_2 and m_1. [Note: The data centers send responses only when they receive a request from the server]

Options :

(C) $$m_2 = \frac{D_2 m_1}{D_1 - 0.2m_1}$$

(D) $$m_2 = \frac{D_2 m_1}{D_1 + 0.2m_1}$$

(E) $$m_2 = \frac{2D_2 m_1}{2D_1 - 0.2m_1}$$

(F) $$m_2 = \frac{2D_2 m_1}{2D_1 + 0.2m_1}$$

Official Answer Key: 6406533776000

MAD1_084 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

A server S requests to retrieve data from two data centers DC_1 and DC_2, which are located at distances of D_1 and D_2 meters respectively from the server S. The speed of data travelling through the medium that connects S with DC_1 is m_1 m/sec and that connects S with DC_2 is m_2 m/sec. With the given media for data transfer between S $\rightarrow DC_1$ and S $\rightarrow DC_2$, it takes an additional 200 milliseconds for the data to reach S from DC_2 than it takes from DC_1 if the complete response is ready only when the data is available to S from both the datacenters. Figure out the value of m_2 (in m/sec) in terms of D_1, D_2 and m_1. [Note: The data centers send responses only when they receive a request from the server]

Options :

(C) $$m_2 = \frac{D_2 m_1}{D_1 - 0.2m_1}$$

(D) $$m_2 = \frac{D_2 m_1}{D_1 + 0.2m_1}$$

(E) $$m_2 = \frac{2D_2 m_1}{2D_1 - 0.2m_1}$$

(F) $$m_2 = \frac{2D_2 m_1}{2D_1 + 0.2m_1}$$

Official Answer Key: 6406533775171

MAD1_091 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz1 May23 .pdf

Let L = {' ', ' ', 'c', 'd', 'A', 'B', 'C', 'D', '0', '1', ' '} be a complete character set (i.e., only these characters can be used to represent text in the document). If a document that uses fixed encoding for all characters is created using the character set L and has a disk size of 2 Kilobytes, the number of characters in the documents would be_____. [Take 1 Byte = 8 bits, 1 KB = 1000 Bytes, 1 MB = 1000 Kilobytes and so on.]

Options :

(B) 2000

(C) 4000

(D) 3000

(E) 16000

Question Number : 108 Question Id : 640653577857 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructior Time : 0

Correct Marks : 3

Official Answer Key: 6406531929660

MAD1_128 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

A certain movie with advertisements is being streamed by the client on the internet with a bandwidth of 5 Mbps. If the total data consumed only in the advertisements is 340 MB and the advertisements comprises 17% of the total duration of the streamed movie, what can be the size of the entire movie (in Megabytes) including the advertisements?

Options :

(F) 20

(7) 340

(8) 2000

(9) 1660

Question Number : 39 Question Id : 640653825024 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a document of 1000 characters written using only the characters of the character set L1 = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j'}. If the character set L1 is to be upgraded by adding all the characters of character set L2 = {'0', '1', '2', '3', '4', '5', '6'} in it, By how many bits will the size of the original document increase?

Options :

Official Answer Key: 6406532774737

MAD1_134 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

Consider a client A that makes a request to server B that fetches the information from Datacenter C which in turn fetches other required information from the Datacenter D. The distance between the client A, server B and datacenters C and D within the network and the speed of information in the medium connecting these nodes is mentioned in the table below.

Nodes	A to B	B to C	C to D
Distance (km)	12000	8500	7500
Speed of Information(m/s)	3×10^8	2×10^8	1.5×10^8

Calculate the round trip latency (in milliseconds) of the network?

Options :

(C) 187

(D) 132.5

(E) 265

(F) 150

Official Answer Key: 6406532774757

MAD1_253 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf

A mobile server and a client initially 750 km apart are moving towards each other in a straight line with the speeds of 165 kmph and 85 kmph respectively. The network bandwidth is set constant to 2 Mbps. Determine how much data (in Gigabytes) is used by the client before the client and the server collide. [speed of light in vacuum: 3 *x* 10^8 m/sec, 1 MB = 1000 KB and so on.]

Options :

(A) 21.6

(B) 5

(C) 2.7

(D) 7.5

Official Answer Key: 6406532239857

MAD1_256 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

A mobile server and a client initially 750 km apart are moving towards each other in a straight line with the speeds of 165 kmph and 85 kmph respectively. The network bandwidth is set constant to 2 Mbps. Determine how much data (in Gigabytes) is used by the client before the client and the server collide. [speed of light in vacuum: 3 *x* 10^8 m/sec, 1 MB = 1000 KB and so on.]

Options :

(A) 21.6

(B) 5

(C) 2.7

(D) 7.5

Official Answer Key: 6406532240953

MAD1_258 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN4 EXAM QPD4 29 Oct.pdf

A mobile server and a client initially 750 km apart are moving towards each other in a straight line with the speeds of 165 kmph and 85 kmph respectively. The network bandwidth is set constant to 2 Mbps. Determine how much data (in Gigabytes) is used by the client before the client and the server collide. [speed of light in vacuum: 3×10^8 m/sec, 1 MB = 1000 KB and so on.]

Options :

(A) 21.6

(B) 5

(C) 2.7

(D) 7.5

Official Answer Key: 6406532241705

MAD1_260 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf, IIT M DIPLOMA AN EXAM QDD3.pdf

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MODERN APPLICATION DEVELOPMENT I (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

(B) YES

(C) NO

Sub-Section Number : 2

(D) 640653146810

Question Shuffling Allowed : Yes

Question Number : 101 Question Id : 640653995532 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Suppose a Gamer wants to buy a Bluetooth earphone which can give him a good gaming experience, for that the delay between the action that happens in his phone and the sound heard by him for that action should be as low as possible so that he can play well. Following this scenario, which of the following option(s) is/are correct?

Options :

(E) He should buy the earphones which have high latency as much possible.

(F) It all depends on Internet Speed, so we can't determine anything.

(7) He should buy the earphones which have low latency as much possible.

(8) He should buy the earphones which have as low Bluetooth version as possible.

Question Number : 102 Question Id : 640653995543 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Official Answer Key: 6406533360886

MAD1_268 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf, IIT M DIPLOMA AN EXAM QDD3.pdf

A certain network connection with a bandwidth of 10 Mbps allows making 1000 requests to a server per second. If each request is modified to have an additional data of 8 Kb, what should be the additional bandwidth required to maintain the rate of 2000 requests per second?

Options :

(E) 18 Mbps

(F) 16 Mbps

(7) 26 Mbps

(8) 36 Mbps

Question Number : 110 Question Id : 640653995541 Question Type : MCQ

Correct Marks : 4.5

Question Label : Multiple Choice Question

Official Answer Key: 6406533360910

MAD1_270 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

A client C and the server S located, 12000 km apart are connected with the help of a cable. A request was sent by the client C to server S, but while sending back the response to the client, the cable breaks and the response is now sent back via air. This change in medium caused an additional delay of 20 msec. As compared to the Round Trip Time of the healthy network, the Round Trip Time (RTT) of the faulty system would _____.

[The speed of light in air is 3×10^8 m/s and that on a cable is 2×10^8 m/s]

Options :

(A) Increase

(B) Decrease

(C) Remain Same

(D) Insufficient Information

Sub-Section Number : 5

(E) 640653146813

Question Shuffling Allowed : Yes

Official Answer Key: 6406533360939

MAD1_284 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

A client C and the server S located, 12000 km apart are connected with the help of a cable. A request was sent by the client C to server S, but while sending back the response to the client, the cable breaks and the response is now sent back via air. This change in medium caused an additional delay of 20 msec. As compared to the Round Trip Time of the healthy network, the Round Trip Time (RTT) of the faulty system would _____.

[The speed of light in air is 3×10^8 m/s and that on a cable is 2×10^8 m/s]

Options :

(A) Increase

(B) Decrease

(C) Remain Same

(D) Insufficient Information

Sub-Section Number : 5

(E) 640653146902

Question Shuffling Allowed : Yes

Official Answer Key: 6406533361723

MAD1_297 | Type: MCQ | Marks: +4.5
Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

A client C and the server S located, 12000 km apart are connected with the help of a cable. A request was sent by the client C to server S, but while sending back the response to the client, the cable breaks and the response is now sent back via air. This change in medium caused an additional delay of 20 msec. As compared to the Round Trip Time of the healthy network, the Round Trip Time (RTT) of the faulty system would _____.

[The speed of light in air is 3×10^8 m/s and that on a cable is 2×10^8 m/s]

Options :

(A) Increase

(B) Decrease

(C) Remain Same

(D) Insufficient Information

Sub-Section Number : 5

(E) 640653146991

Question Shuffling Allowed : Yes

Official Answer Key: 6406533362507

• 1.3 Character Encoding & Representation (ASCII, Unicode, UTF-8 & Binary) (6 Questions)

MAD1_018 | Type: MCQ | Marks: +2.0
Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

The hexadecimal representation of the number 5457_8 is _____ .

Options :

(D) A2F

(E) 31F

(F) 32F

(7) C1E

Official Answer Key: 6406531148353

MAD1_031 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz 1 Jan23 .pdf

A text document of size 2000 bytes. Each character in the document are 8 bits ASCII encoding. What is the number of characters in the text document?

[Note: 8 bits = 1 byte]

Options :

(A) 250

(B) 1000

(C) 2000

(D) 3000

Sub-Section Number : 3

(E) 64065367681

Question Shuffling Allowed : Yes

Is Section Default? : null

Official Answer Key: 6406531562032

MAD1_057 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

Consider two fictitious addressing systems, IPv5 and IPv8, used to store the IP address of a machine. Following are the forms of storing addresses for each system.

IPv5: It is a 45-bit long address denoted by 5 sets of 3-digit octal numbers separated by a hyphen **(-)**.
e.g. `xxx - xxx - xxx - xxx - xxx`

IPv8: It is a 64-bit long address denoted by 4 sets of 4-bit hexadecimal numbers separated by colon **(:)**
e.g. `xxxx:xxxx:xxxx:xxxx`

How would an IPv5 address, `145-123-010-235-453` be represented in IPv8 format based on information given above?

Options :

(A) `0000:3B2B:9821:0652`

(B) `3B2B:9821:0652:0000`

(C) `0000:0652:9821:3B2B`

(D) `0652:9821:3B2B:0000`

Official Answer Key: 6406533774190

MAD1_071 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Consider two fictitious addressing systems, IPv5 and IPv8, used to store the IP address of a machine. Following are the forms of storing addresses for each system.

IPv5: It is a 45-bit long address denoted by 5 sets of 3-digit octal numbers separated by a hyphen **(-)**.
e.g. `xxx - xxx - xxx - xxx - xxx`

IPv8: It is a 64-bit long address denoted by 4 sets of 4-bit hexadecimal numbers separated by colon **(:)**
e.g. `xxxx:xxxx:xxxx:xxxx`

How would an IPv5 address, `145-123-010-235-453` be represented in IPv8 format based on information given above?

Options :

(A) `0000:3B2B:9821:0652`

(B) `3B2B:9821:0652:0000`

(C) `0000:0652:9821:3B2B`

(D) `0652:9821:3B2B:0000`

Official Answer Key: 6406533776003

MAD1_085 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

Consider two fictitious addressing systems, IPv5 and IPv8, used to store the IP address of a machine. Following are the forms of storing addresses for each system.

IPv5: It is a 45-bit long address denoted by 5 sets of 3-digit octal numbers separated by a hyphen **(-)**.
e.g. `xxx - xxx - xxx - xxx - xxx`

IPv8: It is a 64-bit long address denoted by 4 sets of 4-bit hexadecimal numbers separated by colon **(:)**
e.g. `xxxx:xxxx:xxxx:xxxx`

How would an IPv5 address, `145-123-010-235-453` be represented in IPv8 format based on information given above?

Options :

(A) `0000:3B2B:9821:0652`

(B) `3B2B:9821:0652:0000`

(C) `0000:0652:9821:3B2B`

(D) `0652:9821:3B2B:0000`

Official Answer Key: 6406533775174

MAD1_250 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

Which of the following flask view functions will return a 404 error for the URL:
http://127.0.0.1:5000/home/student1/2001 ?

Options :

(A)
```
@app.route('/home/<student_id>/<course_id>')
def home(student_id, course_id):
      details = { 'student_id': student_id, 'course_id': course_id}
```

(B)
```
@app.route('/home/<string:student_id>/<int:course_id>')
def home(student_id, course_id):
      details = { 'student_id': student_id, 'course_id': course_id}
```

(C)
```
@app.route('/home/<string:student_id>/<string:course_id>')
def home(student_id, course_id):
      details = { 'student_id': student_id, 'course_id': course_id}
```

(D)
```
@app.route('/home/student/<student_id>/<course_id>')
def home(student_id, course_id):
      details = { 'student_id': student_id, 'course_id': course_id}
```

Question Number : 108 Question Id : 640653668553 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructior Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a character set L, which consists of only those characters which are used in the statement, "pack my box with five dozen liquor jugs". If this statement is to be saved in a document with minimum encoding, what will be the size of the document given that no other information or context is to be saved?

Options :

(A) 160 bits

(B) 320 bits

(C) 195 bits

(D) 390 bits

Official Answer Key: 6406532239853

• 1.4 HTTP Protocol, URLs, Request/Response Headers & Status Codes (17 Questions)

MAD1_032 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz 1 Jan23 .pdf

Ram wants to do the following operations on his bank website through the browser.

Find the HTTP methods conventionally used for the following operations?

1. To create a new fixed deposit account
2. To read his transaction history
3. To update the account after the transaction of money.
4. To remove his unused account.

Options :

(F) 1 - GET, 2- POST, 3 - PUT, 4 - DELETE

(7) 1 - POST, 2- GET, 3 - PUT, 4 - PATCH

(8) 1 - POST, 2- GET, 3 - PUT, 4 - DELETE

(9) 1 - PUT, 2- GET, 3 - POST, 4 - DELETE

Official Answer Key: 6406531561980

MAD1_046 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

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(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

(B) YES

(C) NO

Sub-Section Number : 2

(D) 640653168508

Question Shuffling Allowed : Yes

Question Number : 67 Question Id : 6406531113759 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is the correct way of calling an external CSS file using HTML?

Options :

(E)

```
<head>
<link url="stylesheet" type="text/css" href="mystyle.css">
</head>
```

(F)

```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

(7)

```
<head>
<link href="stylesheet" type="text/css" file="mystyle.css">
</head>
```

Official Answer Key: 6406533774141

MAD1_047 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

Which command would you use to get **detailed information** about the HTTP request and response, **including headers**, and to **help debug the HTTP request** when using curl?

Options :

(B) `curl -i <URL>`

(C) `curl -v <URL>`

(D) `curl --headers <URL>`

(E) `curl -X GET <URL>`

Question Number : 69 Question Id : 6406531113761 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Read the statements given below carefully and select the correct option.

Statement 1: If an element that belongs to two different classes is styled externally using both the classes, then for the same attribute, it will acquire styling from the latest class in order.
Statement 2: If an element having an ID and a class is styled externally using both its ID and the class, then for the same attribute, it will acquire styling from the ID selector.

Options :

(F) Both statements 1 and 2 are correct

(7) Both statements 1 and 2 are incorrect

(8) Statement 1 is correct but statement 2 is incorrect

(9) Statement 2 is correct but statement 1 is incorrect

Official Answer Key: 6406533774145

MAD1_061 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

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(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

(B) YES

(C) NO

Sub-Section Number : 2

(D) 640653168712

Question Shuffling Allowed : Yes

Question Number : 67 Question Id : 6406531114386 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is the correct way of calling an external CSS file using HTML?

Options :

(E)

```
<head>
<link url="stylesheet" type="text/css" href="mystyle.css">
</head>
```

(F)

```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

(7)

```
<head>
<link href="stylesheet" type="text/css" file="mystyle.css">
</head>
```

Official Answer Key: 6406533775954

MAD1_062 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Which command would you use to get **detailed information** about the HTTP request and response, **including headers**, and to **help debug the HTTP request** when using curl?

Options :

(B) `curl -i <URL>`

(C) `curl -v <URL>`

(D) `curl --headers <URL>`

(E) `curl -X GET <URL>`

Question Number : 69 Question Id : 6406531114388 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Read the statements given below carefully and select the correct option.

Statement 1: If an element that belongs to two different classes is styled externally using both the classes, then for the same attribute, it will acquire styling from the latest class in order.
Statement 2: If an element having an ID and a class is styled externally using both its ID and the class, then for the same attribute, it will acquire styling from the ID selector.

Options :

(F) Both statements 1 and 2 are correct

(7) Both statements 1 and 2 are incorrect

(8) Statement 1 is correct but statement 2 is incorrect

(9) Statement 2 is correct but statement 1 is incorrect

Official Answer Key: 6406533775958

MAD1_075 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MODERN APPLICATION DEVELOPMENT I (COMPUTER BASED EXAM)"

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CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

(B) YES

(C) NO

Sub-Section Number : 2

(D) 640653168613

Question Shuffling Allowed : Yes

Question Number : 67 Question Id : 6406531114099 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is the correct way of calling an external CSS file using HTML?

Options :

(E)

```
<head>
<link url="stylesheet" type="text/css" href="mystyle.css">
</head>
```

(F)

```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

(7)

```
<head>
<link href="stylesheet" type="text/css" file="mystyle.css">
</head>
```

Official Answer Key: 6406533775125

MAD1_076 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

Which command would you use to get **detailed information** about the HTTP request and response, **including headers**, and to **help debug the HTTP request** when using curl?

Options :

(B) `curl -i <URL>`

(C) `curl -v <URL>`

(D) `curl --headers <URL>`

(E) `curl -X GET <URL>`

Question Number : 69 Question Id : 6406531114101 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Read the statements given below carefully and select the correct option.

Statement 1: If an element that belongs to two different classes is styled externally using both the classes, then for the same attribute, it will acquire styling from the latest class in order.
Statement 2: If an element having an ID and a class is styled externally using both its ID and the class, then for the same attribute, it will acquire styling from the ID selector.

Options :

(F) Both statements 1 and 2 are correct

(7) Both statements 1 and 2 are incorrect

(8) Statement 1 is correct but statement 2 is incorrect

(9) Statement 2 is correct but statement 1 is incorrect

Official Answer Key: 6406533775129

MAD1_089 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz1 May23 .pdf

Which of the following statements is/are true?

Options :

(D) HTTP headers (for example “Content-Type”) are case-sensitive

(E) The body/payload in an HTTP request is mandatory

(F) The “Content-Length” header indicates the size of the request body in bytes

(7) All of these

Question Number : 104 Question Id : 640653577862 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructior

Official Answer Key: 6406531929657

MAD1_125 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MODERN APPLICATION DEVELOPMENT I (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
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Options :

(B) YES

(C) NO

Sub-Section Number : 2

(D) 640653122720

Question Shuffling Allowed : Yes

Question Number : 34 Question Id : 640653825020 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Which of the following is not a correct perceivable accessibility principle?
Options :

(E) Provide text alternatives for non-text content.

(F) Provide captions and other alternatives for multimedia.

(7) Make it easier for users to see and hear content.

(8) Give users enough time to read and use content.

Question Number : 35 Question Id : 640653825021 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Which of the below HTTP status codes represents the client error?
Options :

(9) 500-599

Official Answer Key: 6406532774731

MAD1_248 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

What does the term "responsive" mean in the context of web design?

Options :

(B) The ability of a webpage to give visual feedback on any kind of user input.

(C) The ability of a webpage to change its color scheme to "night mode".

(D) The ability of a webpage to alter its layout and appearance to suit different olutions, etc.

(E) All of these.

Question Number : 103 Question Id : 640653668543 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructior Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Match the following HTTP methods in Column A with their conventional usage in Column B.

	Column A		Column B
1	GET	a	Send data to a server to create/update a resource
2	HEAD	b	Update a resource
3	PUT	c	method, generally used to request a resource from the server.
4	POST	d	Asks for a response identical to that of a GET request, but without the response body

Options :

(A) 1-c, 2-d, 3-b, 4-a

(B) 1-c, 2-b, 3-d, 4-a

(C) 1-c, 2-d, 3-a, 4-b

(D) 1-a, 2-d, 3-b, 4-c

Sub-Section Number : 3

(E) 64065395166

Question Shuffling Allowed : Yes

Is Section Default? : null

Official Answer Key: 6406532239811

MAD1_261 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

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(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

(B) YES

(C) NO

Sub-Section Number : 2

(D) 640653146810

Question Shuffling Allowed : Yes

Question Number : 101 Question Id : 640653995532 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Suppose a Gamer wants to buy a Bluetooth earphone which can give him a good gaming experience, for that the delay between the action that happens in his phone and the sound heard by him for that action should be as low as possible so that he can play well. Following this scenario, which of the following option(s) is/are correct?

Options :

(E) He should buy the earphones which have high latency as much possible.

(F) It all depends on Internet Speed, so we can't determine anything.

(7) He should buy the earphones which have low latency as much possible.

(8) He should buy the earphones which have as low Bluetooth version as possible.

Question Number : 102 Question Id : 640653995543 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Consider the below URL:

https://www.google.com/search?q=physics+books

Select the correct answer that identifies the different components of the given URL.

Options :

(A)
Host: http
Protocol: google.com/search
Query string: q=physics+books

(B)
Protocol: http
Host: google.com/search
Query string: q=physics+books

(C)
Protocol: http
Query string: google.com/search
Protocol: q=physics+books

(D)
Host: http
Query string: google.com/search
Protocol: q=physics+books

Sub-Section Number : 3

(E) 640653146811

Question Shuffling Allowed : Yes

Official Answer Key: 6406533360926

MAD1_265 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

The directory listing of a folder and the content of each of its files are given below.

D..ectory l sting:

```
my_dir-|
       |-- base.html
       |-- home.html
       |-- index.html
```

File ame: base.htr

```
<body>
    <h1>Welcome to MAD-I course</h1>
</body>
```

Filename: ome..itml

```
<body>
    <h1>Welcome to PDSA course</h1>
</body>
```

File ame: index.html

```
<body>
    <h1>Welcome to DBMS course</h1>
</body>
```

.f a sir ple P\"hon HTTP server is created for this directory, what will be rendered on the br. wser or bɛ .e URL?

Options :

(A) **Welcome to PDSA course**

(B) **Welcome to DBMS course**

(C) **Welcome to MAD-I course**

(D) **Directory listing for /**

Official Answer Key: 6406533360913

MAD1_271 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf, IIT M DIPLOMA AN EXAM QDD3.pdf

Consider the below server response

HTTP/2.0 500 Internal Server Error

Which of the following is/are **True**?

Options :

(F) HTTP/2.0 is the protocol and version respectively

(7) 500 is the port number

(8) 'Internal Server Error" is the status message

(9) 500 is the status code

(10) HTTP/2.0 is the server failure

Question Number : 113 Question Id : 640653995545 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Official Answer Key: 6406533360920, 6406533360922, 6406533360923

MAD1_276 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MODERN APPLICATION DEVELOPMENT I (COMPUTER BASED EXAM)"

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Options :

(B) YES

(C) NO

Sub-Section Number : 2

(D) 640653146899

Question Shuffling Allowed : Yes

Question Number : 101 Question Id : 640653995814 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Suppose a Gamer wants to buy a Bluetooth earphone which can give him a good gaming experience, for that the delay between the action that happens in his phone and the sound heard by him for that action should be as low as possible so that he can play well. Following this scenario, which of the following option(s) is/are correct?

Options :

(E) He should buy the earphones which have high latency as much possible.

(F) It all depends on Internet Speed, so we can't determine anything.

(7) He should buy the earphones which have low latency as much possible.

(8) He should buy the earphones which have as low Bluetooth version as possible.

Question Number : 102 Question Id : 640653995825 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Consider the below URL:

https://www.google.com/search?q=physics+books

Select the correct answer that identifies the different components of the given URL.

Options :

(A)
Host: http
Protocol: google.com/search
Query string: q=physics+books

(B)
Protocol: http
Host: google.com/search
Query string: q=physics+books

(C)
Protocol: http
Query string: google.com/search
Protocol: q=physics+books

(D)
Host: http
Query string: google.com/search
Protocol: q=physics+books

Sub-Section Number : 3

(E) 640653146900

Question Shuffling Allowed : Yes

Official Answer Key: 6406533361710

MAD1_280 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

The directory listing of a folder and the content of each of its files are given below.

D..ectory l sting:

```
my_dir-|
       |-- base.html
       |-- home.html
       |-- index.html
```

File ame: base.htr

```
<body>
    <h1>Welcome to MAD-I course</h1>
</body>
```

Filename: ome..itml

```
<body>
    <h1>Welcome to PDSA course</h1>
</body>
```

File ame: index.html

```
<body>
    <h1>Welcome to DBMS course</h1>
</body>
```

.f a sir ple P\"hon HTTP server is created for this directory, what will be rendered on the br. wser or bɛ .e URL?

Options :

(A) **Welcome to PDSA course**

(B) **Welcome to DBMS course**

(C) **Welcome to MAD-I course**

(D) **Directory listing for /**

Official Answer Key: 6406533361697

MAD1_289 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MODERN APPLICATION DEVELOPMENT I (COMPUTER BASED EXAM)"

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(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

(B) YES

(C) NO

Sub-Section Number : 2

(D) 640653146988

Question Shuffling Allowed : Yes

Question Number : 101 Question Id : 640653996096 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Suppose a Gamer wants to buy a Bluetooth earphone which can give him a good gaming experience, for that the delay between the action that happens in his phone and the sound heard by him for that action should be as low as possible so that he can play well. Following this scenario, which of the following option(s) is/are correct?

Options :

(E) He should buy the earphones which have high latency as much possible.

(F) It all depends on Internet Speed, so we can't determine anything.

(7) He should buy the earphones which have low latency as much possible.

(8) He should buy the earphones which have as low Bluetooth version as possible.

Question Number : 102 Question Id : 640653996107 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Consider the below URL:

https://www.google.com/search?q=physics+books

Select the correct answer that identifies the different components of the given URL.

Options :

(A)
Host: http
Protocol: google.com/search
Query string: q=physics+books

(B)
Protocol: http
Host: google.com/search
Query string: q=physics+books

(C)
Protocol: http
Query string: google.com/search
Protocol: q=physics+books

(D)
Host: http
Query string: google.com/search
Protocol: q=physics+books

Sub-Section Number : 3

(E) 640653146989

Question Shuffling Allowed : Yes

Official Answer Key: 6406533362494

MAD1_293 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

The directory listing of a folder and the content of each of its files are given below.

D..ectory l sting:

```
my_dir-|
       |-- base.html
       |-- home.html
       |-- index.html
```

File ame: base.htr

```
<body>
    <h1>Welcome to MAD-I course</h1>
</body>
```

Filename: ome..itml

```
<body>
    <h1>Welcome to PDSA course</h1>
</body>
```

File ame: index.html

```
<body>
    <h1>Welcome to DBMS course</h1>
</body>
```

.f a sir ple P\"hon HTTP server is created for this directory, what will be rendered on the br. wser or bɛ .e URL?

Options :

(A) **Welcome to PDSA course**

(B) **Welcome to DBMS course**

(C) **Welcome to MAD-I course**

(D) **Directory listing for /**

Official Answer Key: 6406533362481

WEEK 2: HTML5, CSS3, BOX MODEL & FORM CONTROLS

• 2.1 HTML5 Elements, Semantic Tags & Document Structure (30 Questions)

MAD1_003 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

Consider the following HTML code snippet.

filename: index.html

```
<!DOCTYPE html>
<html lang=" n >
<head>
    <title>CSS Document</title>
    <style>
       .container p.highlight { background-color: red; }
       .content .special { background-color: blue; }
       span.highlight { background-color: green; }
    </style>
</head>
<body>
    <div class="container">
        <p class="highlight">Welcome to our site</p>
        <div class="content">
            <p>This is regular content</p>
            <p class="highlight special">Important information</p>
            <span class="highlight">Notice</span>
        </div>
        <p class="highlight">Footer text</p>
    </div>
    <p class='  hlight">Welcome to our site</p>
</body>
</html>
```

'Vhic" elements from the above document will have a red background color?

Options :

(A) All elements with the class "highlight".

(B) Only the <p> elements that are direct children of .container and have the class "highlight"

(C) All <p> elements with the class "highlight" that are descendants of .container

(D) Only the first <p> element with class "highlight"

Official Answer Key: 6406534366189

MAD1_019 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

The hexadecimal representation of the number 5457_8 is _____ .

Options :

(D) A2F

(E) 31F

(F) 32F

(7) C1E

Official Answer Key: 6406531148367

MAD1_020 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

Consider the following HTML code below.

```
<!DOCTYPE ht l>
<html lang="en">
  <head>
    <meta charset="UTF-8"/>
    <style>
      bo_y{text-align: center}
      p{font-size: 30px;font-style: italic;color: black;}
      .blue{color: green;}
      .red{color: blue;}
      .green{color: red;}
      #myId{color: grey;}
    </style>
  </ ead>
  <bo_y>
    <div>
      <h2>Welcome to IIT</h2>
      <p class="blue red green">Paragraph 1</p>
      <p clas ="red blue green" id="myId">Paragraph 2</p>
      <p clas ="green red blue">Paragraph 3 </p>
    </div>
  </body>
</html>
```

How will the browser render the above given HTML file?

Options :

(A)

Official Answer Key: 6406531148335

MAD1_021 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf, Quiz1 May23 .pdf

Consider the following Python code snippet.

```
filenan  :  ( ode.py

i  ] ort sys
my_ar gs = sys.argv

pri  .  The python file name is:", my_args[1])
pri  .  f"M, username is: {my_args[3]}{my_args[4]}")
```

What v... be t e output on console if the command given is:

```
pyt ion code.py output.py Appdev Michael 2003 1003
```

Options :

(A)
```
The python file name is: code.py
My username is: Michael1003
```

(B)
```
The python file name is: code.py
My username is: Michael2003
```

(C)
```
The python file name is: output.py
My username is: Michael1003
```

(D)
```
The python file name is: output.py
My username is: Michael2003
```

Official Answer Key: 6406531148346

MAD1_022 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf, Quiz 1 Jan23 .pdf

Annie and Curly took Database and Networks courses. Bella took Compiler and Python courses and Deva took Compiler and Database courses. Which of the following list of tuples correctly represents the relationship between names and courses.
names = {0: 'Annie', 1: 'Bella', 2: 'Curly', 3: 'Deva'}
courses = {0: 'Database', 1: 'Compiler', 2: 'Networks', 3: 'Python'}

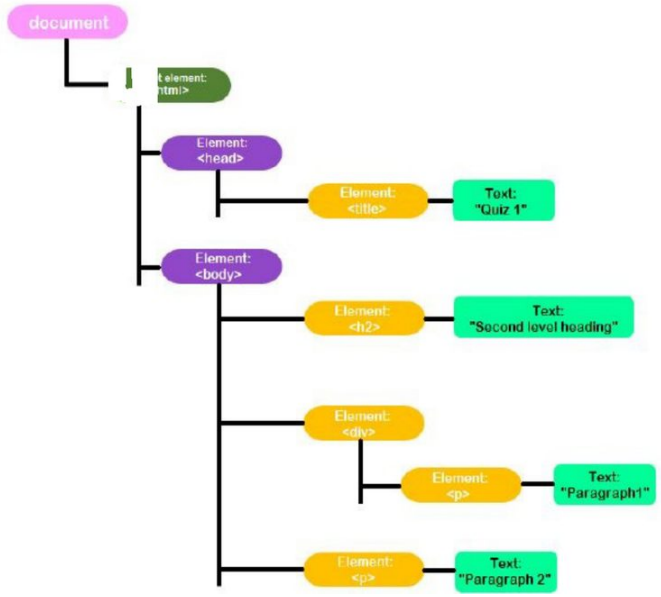
Options :

Official Answer Key: 6406531148355

MAD1_026 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

Consider the DOM structure given below.



Official Answer Key: 6406531148386

MAD1_027 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

Consider a client which is located 6000 kilometers from the server makes a request through the cable. Suddenly after the request reaches the server, the cable breaks and the response is now to be sent to the client via air with the help of repeaters which added a delay of 75 milliseconds. How long will the client have to wait before receiving the response? [**Note:** the speed of light on cable is 2×10^8 m/sec and that in air is 3×10^8 m/sec.]

Options :

(A) 50 milliseconds

(B) 135 milliseconds

(C) 125 milliseconds

(D) 115 milliseconds

MLF

(E)	64065321859
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	15
Number of Questions to be attempted :	15
Section Marks :	50
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
(F)	64065348776
Question Shuffling Allowed :	No

Official Answer Key: 6406531148393

MAD1_029 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz 1 Jan23 .pdf

Match the component in column A with its correct feature given in column B.

Column A	Column B
1. Telephone Networks	a. Wires occupied only when data to be sent
2. Packet Switched Networks	b. First public packet switched network
3. ARPANET	c. Physical wires tied up for duration of call even if nothing said
4. TCP	d. Establishes reliable communications

Options :

(A) 1 - a, 2 - b, 3 - c, 4 - d

(B) 1 - b, 2 - d, 3 - a, 4 - c

(C) 1 - d, 2 - c, 3 - b, 4 - a

(D) 1 - c, 2 - a, 3 - b, 4 - d

Official Answer Key: 6406531562008

MAD1_030 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz 1 Jan23 .pdf

Which of the following option represents a one-to-many relationship in the Entity-Relationship Diagram?

Options :

(E)

Official Answer Key: 6406531562029

MAD1_033 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz 1 Jan23 .pdf

How to represent the number 111_{10} in the binary format?

Options :

(A) 1110000

(B) 1101111

(C) 1100000

(D) 1111000

Official Answer Key: 6406531561987

MAD1_038 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024.pdf, IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024.pdf

Consider the following example.

```
<img src="example.jpg" alt=" Gajendra Circle, IITM Campus">
```

Which of the following accessibility principle is followed in this example?

Options :

(A) Perceivable

(B) Operable

(C) Understandable

(D) Robust

Official Answer Key: 6406532467910

MAD1_040 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024.pdf

Consider the following statements and choose the correct options.

Statement I: An author can write multiple books, but each book is written by only one author.

Statement II: A publisher can publish multiple books, but each book is published by only one publisher.

Statement III: A book belongs to one genre, but a genre can have multiple books.

Options :

(B) Statement I represents a many-to-many relationship between Author and

(C) Statement I represents a one-to-many relationship between Author and Book.

(D) Statement II represents a one-to-many relationship between Publisher and

(E) Statement III represents a many-to-one relationship between Book and Genre.

Official Answer Key: 6406532467935, 6406532467936, 6406532467937

MAD1_042 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024.pdf

Consider the following statements and choose the correct options.

Statement I: An author can write multiple books, but each book is written by only one author.

Statement II: A publisher can publish multiple books, but each book is published by only one publisher.

Statement III: A book belongs to one genre, but a genre can have multiple books.

Options :

(B) Statement I represents a many-to-many relationship between Author and

(C) Statement I represents a one-to-many relationship between Author and Book.

(D) Statement II represents a one-to-many relationship between Publisher and

(E) Statement III represents a many-to-one relationship between Book and Genre.

Official Answer Key: 6406532468759, 6406532468760, 6406532468761

MAD1_044 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024.pdf

Consider the following statements and choose the correct options.

Statement I: An author can write multiple books, but each book is written by only one author.

Statement II: A publisher can publish multiple books, but each book is published by only one publisher.

Statement III: A book belongs to one genre, but a genre can have multiple books.

Options :

(B) Statement I represents a many-to-many relationship between Author and

(C) Statement I represents a one-to-many relationship between Author and Book.

(D) Statement II represents a one-to-many relationship between Publisher and

(E) Statement III represents a many-to-one relationship between Book and Genre.

Official Answer Key: 6406532469583, 6406532469584, 6406532469585

MAD1_050 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf, IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Consider the following HTML code:

```
<!DOCTYPE html>
<html lang=    >
<head>
   <title>Document</title>
   <style>
       div {
           color: blue;
       }
           {
           color: yellow;
       }
       .s2{
           color: green;
       }
       h3{
           color: red;
       }
   </style>
</head>
<body>
   <div class=  2" id="s1"><h3>IITM Online BS Degree Program</h3></div>
</body>
</html>
```

What is the colour of the printed text "IITM Online BS Degree Program"?

Options :

(A) BLUE

(B) YELLOW

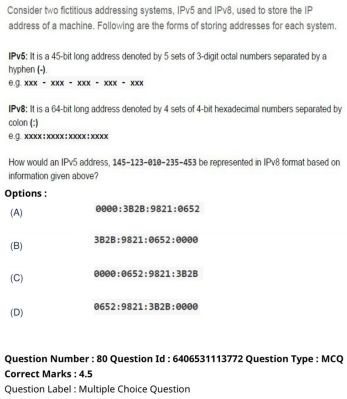
(C) GREEN

(D) RED

Official Answer Key: 6406533774163

MAD1_058 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf



Consider two fictitious addressing systems, IPv5 and IPv8, used to store the IP address of a machine. Following are the forms of storing addresses for each system.

IPv5: It is a 45-bit long address denoted by 5 sets of 3-digit octal numbers separated by a hyphen (-).
e.g. `xxx - xxx - xxx - xxx - xxx`

IPv8: It is a 64-bit long address denoted by 4 sets of 4-bit hexadecimal numbers separated by colon (:)
e.g. `xxxx:xxxx:xxxx:xxxx`

How would an IPv5 address, `145-123-010-235-453` be represented in IPv8 format based on information given above?

Options :

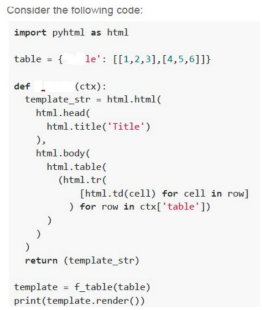
(A) `0000:3B2B:9821:0652`

(B) `3B2B:9821:0652:0000`

(C) `0000:0652:9821:3B2B`

(D) `0652:9821:3B2B:0000`

Question Number : 80 Question Id : 6406531113772 Question Type : MCQ Correct Marks : 4.5
Question Label : Multiple Choice Question



Consider the following code:

```
import pyhtml as html

table = {    le': [[1,2,3],[4,5,6]]}

def _      (ctx):
  template_str = html.html(
    html.head(
      html.title('Title')
    ),
    html.body(
      html.table(
        (html.tr(
            [html.td(cell) for cell in row]
          ) for row in ctx['table'])
      )
    )
  )
  return (template_str)

template = f_table(table)
print(template.render())
```

Official Answer Key: 6406533774192

MAD1_072 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Consider two fictitious addressing systems, IPv5 and IPv8, used to store the IP address of a machine. Following are the forms of storing addresses for each system.

IPv5: It is a 45-bit long address denoted by 5 sets of 3-digit octal numbers separated by a hyphen (-).
e.g. `xxx - xxx - xxx - xxx - xxx`

IPv8: It is a 64-bit long address denoted by 4 sets of 4-bit hexadecimal numbers separated by colon (:)
e.g. `xxxx:xxxx:xxxx:xxxx`

How would an IPv5 address, `145-123-010-235-453` be represented in IPv8 format based on information given above?

Options :

(A) `0000:3B2B:9821:0652`

(B) `3B2B:9821:0652:0000`

(C) `0000:0652:9821:3B2B`

(D) `0652:9821:3B2B:0000`

Question Number : 80 Question Id : 6406531114399 Question Type : MCQ Correct Marks : 4.5
Question Label : Multiple Choice Question

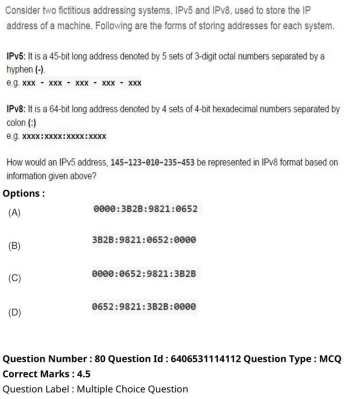
Consider the following code:

```
import pyhtml as html

table = {    le': [[1,2,3],[4,5,6]]}

def _      (ctx):
  template_str = html.html(
    html.head(
      html.title('Title')
    ),
    html.body(
      html.table(
        (html.tr(
            [html.td(cell) for cell in row]
          ) for row in ctx['table'])
      )
    )
  )
  return (template_str)

template = f_table(table)
print(template.render())
```

Official Answer Key: 6406533776005

MAD1_086 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf



Consider two fictitious addressing systems, IPv5 and IPv8, used to store the IP address of a machine. Following are the forms of storing addresses for each system.

IPv5: It is a 45-bit long address denoted by 5 sets of 3-digit octal numbers separated by a hyphen (-).
e.g. `xxx - xxx - xxx - xxx - xxx`

IPv8: It is a 64-bit long address denoted by 4 sets of 4-bit hexadecimal numbers separated by colon (:)
e.g. `xxxx:xxxx:xxxx:xxxx`

How would an IPv5 address, `145-123-010-235-453` be represented in IPv8 format based on information given above?

Options :

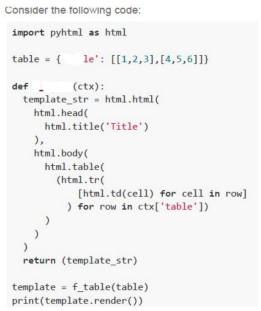
(A) `0000:3B2B:9821:0652`

(B) `3B2B:9821:0652:0000`

(C) `0000:0652:9821:3B2B`

(D) `0652:9821:3B2B:0000`

Question Number : 80 Question Id : 6406531114112 Question Type : MCQ Correct Marks : 4.5
Question Label : Multiple Choice Question



Consider the following code:

```
import pyhtml as html

table = {    le': [[1,2,3],[4,5,6]]}

def _      (ctx):
  template_str = html.html(
    html.head(
      html.title('Title')
    ),
    html.body(
      html.table(
        (html.tr(
            [html.td(cell) for cell in row]
          ) for row in ctx['table'])
      )
    )
  )
  return (template_str)

template = f_table(table)
print(template.render())
```

Official Answer Key: 6406533775176

MAD1_096 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

Consider the following HTML code snippet.

filename: index.html

```
<!DOCTYPE html>
<html lang=" n >
<head>
    <title>CSS Document</title>
    <style>
       .container p.highlight { background-color: red; }
       .content .special { background-color: blue; }
       span.highlight { background-color: green; }
    </style>
</head>
<body>
    <div class="container">
        <p class="highlight">Welcome to our site</p>
        <div class="content">
            <p>This is regular content</p>
            <p class="highlight special">Important information</p>
            <span class="highlight">Notice</span>
        </div>
        <p class="highlight">Footer text</p>
    </div>
    <p class="  hlight">Welcome to our site</p>
</body>
</html>
```

'Vhic elements from the above document will have a red background color?

Options :

(A) All elements with the class "highlight".

(B) Only the <p> elements that are direct children of .container and have the class "highlight"

(C) All <p> elements with the class "highlight" that are descendants of .container

(D) Only the first <p> element with class "highlight"

Official Answer Key: 6406534364647

MAD1_111 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

Consider the following HTML code snippet.

filename: index.html

```
<!DOCTYPE html>
<html lang=" n >
<head>
    <title>CSS Document</title>
    <style>
       .container p.highlight { background-color: red; }
       .content .special { background-color: blue; }
       span.highlight { background-color: green; }
    </style>
</head>
<body>
    <div class="container">
        <p class="highlight">Welcome to our site</p>
        <div class="content">
            <p>This is regular content</p>
            <p class="highlight special">Important information</p>
            <span class="highlight">Notice</span>
        </div>
        <p class="highlight">Footer text</p>
    </div>
    <p class='  ` hlight">Welcome to our site</p>
</body>
</html>
```

'Vhic elements from the above document will have a red background color?

Options :

(A) All elements with the class "highlight".

(B) Only the <p> elements that are direct children of .container and have the class "highlight"

(C) All <p> elements with the class "highlight" that are descendants of .container

(D) Only the first <p> element with class "highlight"

Official Answer Key: 6406534365418

MAD1_124 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MODERN APPLICATION DEVELOPMENT I (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

(B) YES

(C) NO

Sub-Section Number : 2

(D) 640653122720

Question Shuffling Allowed : Yes

Question Number : 34 Question Id : 640653825020 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Which of the following is not a correct perceivable accessibility principle?
Options :

(E) Provide text alternatives for non-text content.

(F) Provide captions and other alternatives for multimedia.

(7) Make it easier for users to see and hear content.

(8) Give users enough time to read and use content.

Question Number : 35 Question Id : 640653825021 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Which of the below HTTP status codes represents the client error?
Options :

(9) 500-599

Official Answer Key: 6406532774729

MAD1_129 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf

A certain movie with advertisements is being streamed by the client on the internet with a bandwidth of 5 Mbps. If the total data consumed only in the advertisements is 340 MB and the advertisements comprises 17% of the total duration of the streamed movie, what can be the size of the entire movie (in Megabytes) including the advertisements?

Options :

(F) 20

(7) 540

(8) 2000

(9) 1660

Question Number : 39 Question Id : 640653825024 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a document of 1000 characters written using only the characters of the character set L1 = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j'}. If the character set L1 is to be upgraded by adding all the characters of character set L2 = {'0', '1', '2', '3', '4', '5', '6'} in it, By how many bits will the size of the original document increase?

Options :

Official Answer Key: 6406532774744

MAD1_130 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

The following Python code snippet generates the output on the terminal.

```
from string import Template
s = Template( $name studies at $university')
b = {      :  Bob'}

out1 = Template('$name studies $subject').safe_substitute(b)
print(out1)

out2 = s.substitute(name='Alice', university='Oxford')
print(out2)

out3 = Template('$name studies $subject').substitute(b)
print(out3)
```

Which of the following is the correct output?

Options :

(E)
```
Bob studies at $subject
Alice studies at Oxford
KeyError: 'subject'
```

(F)
```
Bob studies $subject
Alice studies at Oxford
KeyError: 'subject'
```

(7)
```
Bob studies $subject
Alice studies at Oxford
Bob studies $subject
```

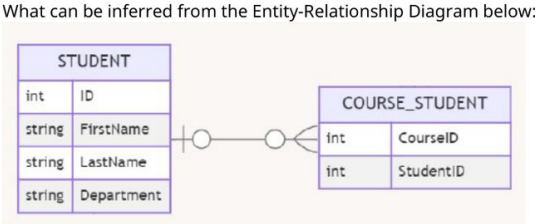
(8)
```
Bob studies
Alice studies at Oxford
KeyError: 'subject'
```

Official Answer Key: 6406532774748

MAD1_136 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

What can be inferred from the Entity-Relationship Diagram below:



Options :

(C) A student can exist without enrolling to any course

(D) A student can have more than one courses

(E) A student needs to have at least one course

(F) A course must belong to one and only one student

Question Number : 48 Question Id : 640653825033 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Consider the following Flask code.

Official Answer Key: 6406532774763, 6406532774764

MAD1_138 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

A certain movie with advertisements is being streamed by the client on the internet with a bandwidth of 5 Mbps. If the total data consumed only in the advertisements is 340 MB and the advertisements comprises 17% of the total duration of the streamed movie, what can be the size of the entire movie (in Megabytes) including the advertisements?

Options :

(F) 20

(7) 540

(8) 2000

(9) 1660

Question Number : 39 Question Id : 640653827493 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a document of 1000 characters written using only the characters of the character set L1 = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j'}. If the character set L1 is to be upgraded by adding all the characters of character set L2 = {'0', '1', '2', '3', '4', '5', '6'} in it, By how many bits will the size of the original document increase?

Options :

Official Answer Key: 6406532781268

MAD1_140 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 7 July 2024.pdf

A certain movie with advertisements is being streamed by the client on the internet with a bandwidth of 5 Mbps. If the total data consumed only in the advertisements is 340 MB and the advertisements comprises 17% of the total duration of the streamed movie, what can be the size of the entire movie (in Megabytes) including the advertisements?

Options :

(F) 20

(7) 540

(8) 2000

(9) 1660

Question Number : 39 Question Id : 640653827781 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a document of 1000 characters written using only the characters of the character set L1 = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j'}. If the character set L1 is to be upgraded by adding all the characters of character set L2 = {'0', '1', '2', '3', '4', '5', '6'} in it, By how many bits will the size of the original document increase?

Options :

Official Answer Key: 6406532782064

MAD1_254 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

A mobile server and a client initially 750 km apart are moving towards each other in a straight line with the speeds of 165 kmph and 85 kmph respectively. The network bandwidth is set constant to 2 Mbps. Determine how much data (in Gigabytes) is used by the client before the client and the server collide. [speed of light in vacuum: 3×10^8 m/sec, 1 MB = 1000 KB and so on.]

Options :

(A) 21.6

(B) 5

(C) 2.7

(D) 7.5

Official Answer Key: 6406532239815, 6406532239817, 6406532239818

MAD1_264 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

The following Python code snippet generates the output on the terminal.

```
from string imp rt Template
greet_tr~plate = Template('Hello, $name! Welcome to $place.')

greeting1 = greet_template.substitute(name='Alice', place='Wonderland')
print(greeting1)

data = dict(na e='Bob')
greeting2 = greet_template.safe_substitute(data)
print(greeting2)

greeting3 = greet_template.substitute(data)
print(greeting3)
```

Which of wie following is the correct output?

Options :

(A)
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
Hello, Bob! Welcome to $place.
```

(B)
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to .
Hello, Bob! Welcome to $place.
```

(C)
```
Hello, Alice! Welcome to Wonderland.
KeyError: 'what'
Hello, Bob! Welcome to $place.
```

(D)
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
KeyError: 'place'
```

Official Answer Key: 6406533360907

MAD1_279 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

The following Python code snippet generates the output on the terminal.

```
from string imp rt Template
greet_tr~plate = Template('Hello, $name! Welcome to $place.')

greeting1 = greet_template.substitute(name='Alice', place='Wonderland')
print(greeting1)

data = dict(na e='Bob')
greeting2 = greet_template.safe_substitute(data)
print(greeting2)

greeting3 = greet_template.substitute(data)
print(greeting3)
```

Which of wie following is the correct output?

Options :

(A)

```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
Hello, Bob! Welcome to $place.
```

(B)

```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to .
Hello, Bob! Welcome to $place.
```

(C)

```
Hello, Alice! Welcome to Wonderland.
KeyError: 'what'
Hello, Bob! Welcome to $place.
```

(D)

```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
KeyError: 'place'
```

Official Answer Key: 6406533361691

MAD1_292 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

The following Python code snippet generates the output on the terminal.

```
from string imp rt Template
greet_tr~plate = Template('Hello, $name! Welcome to $place.')

greeting1 = greet_template.substitute(name='Alice', place='Wonderland')
print(greeting1)

data = dict(na e='Bob')
greeting2 = greet_template.safe_substitute(data)
print(greeting2)

greeting3 = greet_template.substitute(data)
print(greeting3)
```

Which of wie following is the correct output?

Options :

(A)
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
Hello, Bob! Welcome to $place.
```

(B)
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to .
Hello, Bob! Welcome to $place.
```

(C)
```
Hello, Alice! Welcome to Wonderland.
KeyError: 'what'
Hello, Bob! Welcome to $place.
```

(D)
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
KeyError: 'place'
```

Official Answer Key: 6406533362475

• 2.2 HTML Forms, Input Types, Attributes & Form Submission (64 Questions)

MAD1_012 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

Consider the following Python code snippet and answer the given subquestions

Filename: webapp.py

```
import sys
from jinja2 import Template

args = sys.argv
frameworks = {
    'djar_o': 'python',
    'react': 'javascript',
    'flask': 'python',
    'vue': 'javascript'
}

languages = {
    'python': 'server-side',
    'javascript': 'client-side'
}

template = Template("{{ framework }} is a {{ lang_type }} {{ language }}
framework.")

if len(args) >= 3 and args[1] in frameworks:
    framework_name = args[1]
    language = frameworks[framework_name]
    if args[2] == 'type' and language in languages:
        lang_type = languages[language]
        print(template.render(framework=framework_name, language=language,
lang_type=lang_type))
    else:
        print("Invalid second argument or language not found!")
else:
    print("Invalid command")
```

Sub questions

Official Answer Key: 6406534366218, 6406534366219

MAD1_024 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

For a network bandwidth of 8 Gbps, what should be the size of each request if 5000 such requests are to be sent over the network per second? [**Use these relations:** 1 Byte = 8 bits, 1 KB = 1000 Bytes, 1 MB = 1000 KBs and so on.]

Options :

(A) 1.6 KB

(B) 200 KB

(C) 1.6 MB

(D) 200 MB

Official Answer Key: 6406531148339, 6406531148341

MAD1_054 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

Based on the provided latency and response size information, which of the following statements is/are correct?

(Assume the speed of light in cable as `2×10^8 m/s` for all the options and `ms` is a millisecond)

Options :

(F) The speed of light in a cable is approximately `2×10^8 m/s`, resulting in a latency of approximately `5 ms`(millisecond) for a 1000 km distance.

(7) For a data centre located 2000 km away, a one-way request will take approximately `10 ms`, to reach the server, assuming the speed of light in the cable.

(8) The round-trip latency for communication with a data centre 2000 km away is approximately `50ms`.

(9) If a response size is `1 KB`, a maximum of 50 requests per second can be handled considering a round-trip latency of `20ms`.

Question Number : 77 Question Id : 6406531113769 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Official Answer Key: 6406533774176, 6406533774177, 6406533774179

MAD1_055 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

Based on the provided latency and response size information, which of the following statements is/are correct?

(Assume the speed of light in cable as `2×10^8 m/s` for all the options and `ms` is a millisecond)

Options :

(F) The speed of light in a cable is approximately `2×10^8 m/s`, resulting in a latency of approximately `5 ms`(millisecond) for a 1000 km distance.

(7) For a data centre located 2000 km away, a one-way request will take approximately `10 ms`, to reach the server, assuming the speed of light in the cable.

(8) The round-trip latency for communication with a data centre 2000 km away is approximately `50ms`.

(9) If a response size is `1 KB`, a maximum of 50 requests per second can be handled considering a round-trip latency of `20ms`.

Question Number : 77 Question Id : 6406531113769 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Official Answer Key: 6406533774181, 6406533774183

MAD1_068 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Based on the provided latency and response size information, which of the following statements is/are correct?

(Assume the speed of light in cable as `2×10^8 m/s` for all the options and `ms` is a millisecond)

Options :

(F) The speed of light in a cable is approximately `2×10^8 m/s`, resulting in a latency of approximately `5 ms`(millisecond) for a 1000 km distance.

(7) For a data centre located 2000 km away, a one-way request will take approximately `10 ms`, to reach the server, assuming the speed of light in the cable.

(8) The round-trip latency for communication with a data centre 2000 km away is approximately `50ms`.

(9) If a response size is `1 KB`, a maximum of 50 requests per second can be handled considering a round-trip latency of `20ms`.

Question Number : 77 Question Id : 6406531114396 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Official Answer Key: 6406533775989, 6406533775990, 6406533775992

MAD1_069 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Based on the provided latency and response size information, which of the following statements is/are correct?

(Assume the speed of light in cable as `2×10^8 m/s` for all the options and `ms` is a millisecond)

Options :

(F) The speed of light in a cable is approximately `2×10^8 m/s`, resulting in a latency of approximately `5 ms`(millisecond) for a 1000 km distance.

(7) For a data centre located 2000 km away, a one-way request will take approximately `10 ms`, to reach the server, assuming the speed of light in the cable.

(8) The round-trip latency for communication with a data centre 2000 km away is approximately `50ms`.

(9) If a response size is `1 KB`, a maximum of 50 requests per second can be handled considering a round-trip latency of `20ms`.

Question Number : 77 Question Id : 6406531114396 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Official Answer Key: 6406533775994, 6406533775996

MAD1_082 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

Based on the provided latency and response size information, which of the following statements is/are correct?

(Assume the speed of light in cable as `2×10^8 m/s` for all the options and `ms` is a millisecond)

Options :

(F) The speed of light in a cable is approximately `2×10^8 m/s`, resulting in a latency of approximately `5 ms`(millisecond) for a 1000 km distance.

(7) For a data centre located 2000 km away, a one-way request will take approximately `10 ms`, to reach the server, assuming the speed of light in the cable.

(8) The round-trip latency for communication with a data centre 2000 km away is approximately `50ms`.

(9) If a response size is `1 KB`, a maximum of 50 requests per second can be handled considering a round-trip latency of `20ms`.

Question Number : 77 Question Id : 6406531114109 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Official Answer Key: 6406533775160, 6406533775161, 6406533775163

MAD1_083 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

Based on the provided latency and response size information, which of the following statements is/are correct?

(Assume the speed of light in cable as `2×10^8 m/s` for all the options and `ms` is a millisecond)

Options :

(F) The speed of light in a cable is approximately `2×10^8 m/s`, resulting in a latency of approximately `5 ms`(millisecond) for a 1000 km distance.

(7) For a data centre located 2000 km away, a one-way request will take approximately `10 ms`, to reach the server, assuming the speed of light in the cable.

(8) The round-trip latency for communication with a data centre 2000 km away is approximately `50ms`.

(9) If a response size is `1 KB`, a maximum of 50 requests per second can be handled considering a round-trip latency of `20ms`.

Question Number : 77 Question Id : 6406531114109 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Official Answer Key: 6406533775165, 6406533775167

MAD1_104 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

Consider the following Python code snippet and answer the given subquestions

Filename: webapp.py

```
import sys
from jinja2 import Template

args = sys.argv
frameworks = {
    'djar_o': 'python',
    'react': 'javascript',
    'flask': 'python',
    'vue': 'javascript'
}

languages = {
    'python': 'server-side',
    'javascript': 'client-side'
}

template = Template("{{ framework }} is a {{ lang_type }} {{ language }}
framework.")

if len(args) >= 3 and args[1] in frameworks:
    framework_name = args[1]
    language = frameworks[framework_name]
    if args[2] == 'type' and language in languages:
        lang_type = languages[language]
        print(template.render(framework=framework_name, language=language,
lang_type=lang_type))
    else:
        print("Invalid second argument or language not found!")
else:
    print("Invalid command")
```

Sub questions

Official Answer Key: 6406534364676, 6406534364677

MAD1_119 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

Consider the following Python code snippet and answer the given subquestions

Filename: webapp.py

```
import sys
from jinja2 import Template

args = sys.argv
frameworks = {
    'djar_o': 'python',
    'react': 'javascript',
    'flask': 'python',
    'vue': 'javascript'
}

languages = {
    'python': 'server-side',
    'javascript': 'client-side'
}

template = Template("{{ framework }} is a {{ lang_type }} {{ language }}
framework.")

if len(args) >= 3 and args[1] in frameworks:
    framework_name = args[1]
    language = frameworks[framework_name]
    if args[2] == 'type' and language in languages:
        lang_type = languages[language]
        print(template.render(framework=framework_name, language=language,
lang_type=lang_type))
    else:
        print("Invalid second argument or language not found!")
else:
    print("Invalid command")
```

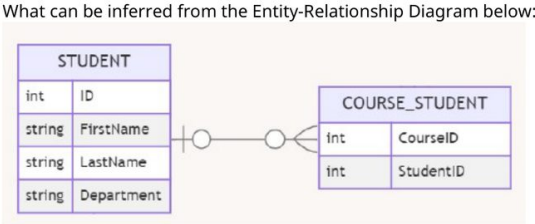
Sub questions

Official Answer Key: 6406534365447, 6406534365448

MAD1_137 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

What can be inferred from the Entity-Relationship Diagram below:



Options :

(C) A student can exist without enrolling to any course

(D) A student can have more than one courses

(E) A student needs to have at least one course

(F) A course must belong to one and only one student

Question Number : 48 Question Id : 640653825033 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Consider the following Flask code.

Official Answer Key: 6406532774779, 6406532774780, 6406532774781

MAD1_143 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf, IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Consider the following statements and select the correct option.

Statement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

Statement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

Options :

(B) Statement 1 is correct, but statement 2 is incorrect.

(C) Statement 2 is correct, but statement 1 is incorrect.

(D) Both statements 1 and 2 are correct

(E) Both statements 1 and 2 are incorrect

Official Answer Key: 6406535115556

MAD1_147 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535115559, 6406535115561, 6406535115562

MAD1_148 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

Options :

(F)

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

Official Answer Key: 6406535115592, 6406535115594

MAD1_151 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 89 Question Id : 6406531527815 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535115584, 6406535115585, 6406535115586

MAD1_152 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 89 Question Id : 6406531527815 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535115588, 6406535115589

MAD1_160 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535116402, 6406535116404, 6406535116405

MAD1_161 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

Options :

(F)

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

Official Answer Key: 6406535116435, 6406535116437

MAD1_164 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 89 Question Id : 6406531528118 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535116427, 6406535116428, 6406535116429

MAD1_165 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 89 Question Id : 6406531528118 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535116431, 6406535116432

MAD1_171 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535118333, 6406535118335, 6406535118336

MAD1_172 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

Options :

(F)

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

Official Answer Key: 6406535118366, 6406535118368

MAD1_175 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 89 Question Id : 6406531528780 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535118358, 6406535118359, 6406535118360

MAD1_176 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 89 Question Id : 6406531528780 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535118362, 6406535118363

MAD1_183 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(C) 1 → B, 2 → C, 3 → D, 4 → A

(D) 1 → A, 2 → B, 3 → C, 4 → D

(E) 1 → B, 2 → D, 3 → C, 4 → A

(F) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535130953, 6406535130955, 6406535130956

MAD1_184 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

Options :

(F)

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

Official Answer Key: 6406535130986, 6406535130988

MAD1_187 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 138 Question Id : 6406531533258 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535130978, 6406535130979, 6406535130980

MAD1_188 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 138 Question Id : 6406531533258 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535130982, 6406535130983

MAD1_195 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(C) 1 → B, 2 → C, 3 → D, 4 → A

(D) 1 → A, 2 → B, 3 → C, 4 → D

(E) 1 → B, 2 → D, 3 → C, 4 → A

(F) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535131849, 6406535131851, 6406535131852

MAD1_196 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

Options :

(F)

```python
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

Official Answer Key: 6406535131882, 6406535131884

MAD1_199 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 138 Question Id : 6406531533583 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535131874, 6406535131875, 6406535131876

MAD1_200 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 138 Question Id : 6406531533583 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535131878, 6406535131879

MAD1_206 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(C) 1 → B, 2 → C, 3 → D, 4 → A

(D) 1 → A, 2 → B, 3 → C, 4 → D

(E) 1 → B, 2 → D, 3 → C, 4 → A

(F) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535132745, 6406535132747, 6406535132748

MAD1_207 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

Options :

(F)

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

Official Answer Key: 6406535132778, 6406535132780

MAD1_210 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 138 Question Id : 6406531533908 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535132770, 6406535132771, 6406535132772

MAD1_211 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 138 Question Id : 6406531533908 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535132774, 6406535132775

MAD1_217 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535125826, 6406535125828, 6406535125829

MAD1_218 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

Options :

(F)

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

Official Answer Key: 6406535125859, 6406535125861

MAD1_221 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 75 Question Id : 6406531531449 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535125851, 6406535125852, 6406535125853

MAD1_222 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 75 Question Id : 6406531531449 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535125855, 6406535125856

MAD1_229 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535128318, 6406535128320, 6406535128321

MAD1_230 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

Options :

(F)

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

Official Answer Key: 6406535128351, 6406535128353

MAD1_233 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 75 Question Id : 6406531532332 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535128343, 6406535128344, 6406535128345

MAD1_234 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 75 Question Id : 6406531532332 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535128347, 6406535128348

MAD1_240 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535129888, 6406535129890, 6406535129891

MAD1_241 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

Options :

(F)

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

Official Answer Key: 6406535129921, 6406535129923

MAD1_244 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 75 Question Id : 6406531532895 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535129913, 6406535129914, 6406535129915

MAD1_245 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Consider the web URL.

http://localhost:5000/

Which of the following is/are **True**?

Options :

(A) http is the protocol.

(B) 5000 is the port number.

(C) The trailing / represents the base URL .

(D) 5000 is the status code.

Question Number : 75 Question Id : 6406531532895 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following HTML snippet:

******

Which of the following is **true**?

Options :

(E) If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

(F) If the image file is missing, browsers display it as a broken image.

(7) The ***alt*** attribute is mandatory in HTML5.

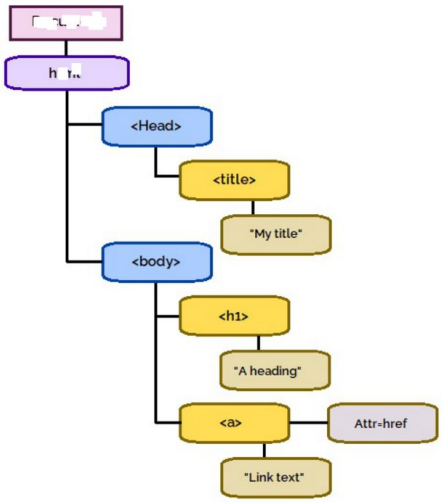
(8) ***Logo.png*** fails to load because the ***<img>*** tag only allows “jpeg/jpg” files.

Official Answer Key: 6406535129917, 6406535129918

MAD1_249 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

Consider the following DOM structure and select the HTML document that correctly represents the DOM structure .



Options :

(A)

```
<!DOCTYPE html>
<head>
    <title>My title</title>
<body>
    <h1>A heading</h1>
    <a href="link">Link text</a>
```

Official Answer Key: 6406532239834

MAD1_262 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

Consider the following HTML code and select the correct rendered output from the following.

```
<!DOCTYPE html>
<html lang=    >
<head>
    <title>Document</title>
</head>
<body>
    <table border="2">
        <tr>
          <th>One</th>
          <td>Four</td>
        </tr>
        <tr>
          <th> Two</th>
          <td> Three</td>
        </tr>
    </table>
</body>
</html>
```

Options :

(A)

One	Four
Two	Three

(B)

One	**Two**
Four	Three

(C)

One	**Two**
Three	Four

(D)

One	**Four**
Two	Three

Official Answer Key: 6406533360888

MAD1_272 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

Consider the below server response

HTTP/2.0 500 Internal Server Error

Which of the following is/are **True**?

Options :

(F) HTTP/2.0 is the protocol and version respectively

(7) 500 is the port number

(8) 'Internal Server Error" is the status message

(9) 500 is the status code

(10) HTTP/2.0 is the server failure

Question Number : 113 Question Id : 640653995545 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Official Answer Key: 6406533360935, 6406533360936

MAD1_273 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

Consider the following code snippet.

```
@app.route('/name/<type>')
def        .       (type):
    # CODE BLOCK HERE
```

Assume the database has a "product" table represented by class "Product" which has a TEXT column . If we want the server to return a 404 status code when a user goes to the route pe>' with a "type" that does not exist in the database, which of the following lines would give us the desired output?

Options :

(F)

Official Answer Key: 6406533360942, 6406533360943, 6406533360944

MAD1_274 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

Suppose the value of **Method** in form.html is GET (form method is set to GET) and in the input box we write "Application", after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

Options :

(A) http://127.0.0.1:5000/?name=Application

(B) http://127.0.0.1:5000/name=Application

(C) http://127.0.0.1:5000/?name

Official Answer Key: 6406533360896

MAD1_275 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

Suppose the value of **Method** in form.html is POST (form method is set to POST) and in the input box we write "Application", after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

Options :

(B) http://127.0.0.1:5000/?name:Application

(C) http://127.0.0.1:5000/?Application

(D) http://127.0.0.1:5000/?name

(E) http://127.0.0.1:5000/

Java

(F)	64065369407
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	16
Number of Questions to be attempted :	16
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0

Official Answer Key: 6406533360903

MAD1_277 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

Consider the following HTML code and select the correct rendered output from the following.

```
<!DOCTYPE html>
<html lang=    >
<head>
    <title>Document</title>
</head>
<body>
    <table border="2">
        <tr>
          <th>One</th>
          <td>Four</td>
        </tr>
        <tr>
          <th> Two</th>
          <td> Three</td>
        </tr>
    </table>
</body>
</html>
```

Options :

(A)

One	Four
Two	Three

(B)

One	**Two**
Four	Three

(C)

One	**Two**
Three	Four

(D)

One	**Four**
Two	Three

Official Answer Key: 6406533361672

MAD1_285 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

Consider the below server response

HTTP/2.0 500 Internal Server Error

Which of the following is/are **True**?

Options :

(F) HTTP/2.0 is the protocol and version respectively

(7) 500 is the port number

(8) 'Internal Server Error" is the status message

(9) 500 is the status code

(10) HTTP/2.0 is the server failure

Question Number : 113 Question Id : 640653995827 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Official Answer Key: 6406533361719, 6406533361720

MAD1_286 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

Consider the following code snippet.

```
@app.route('/name/<type>')
def      .      (type):
    # CODE BLOCK HERE
```

Assume the database has a "product" table represented by class "Product" which has a TEXT column . If we want the server to return a 404 status code when a user goes to the route pe>' with a "type" that does not exist in the database, which of the following lines would give us the desired output?

Options :

(F)

Official Answer Key: 6406533361726, 6406533361727, 6406533361728

MAD1_287 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

Suppose the value of **Method** in form.html is GET (form method is set to GET) and in the input box we write "Application", after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

Options :

(A) http://127.0.0.1:5000/?name=Application

(B) http://127.0.0.1:5000/name=Application

(C) http://127.0.0.1:5000/?name

Official Answer Key: 6406533361680

MAD1_288 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

Suppose the value of **Method** in form.html is POST (form method is set to POST) and in the input box we write "Application", after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

Options :

(B) http://127.0.0.1:5000/?name:Application

(C) http://127.0.0.1:5000/?Application

(D) http://127.0.0.1:5000/?name

(E) http://127.0.0.1:5000/

Java

(F)	64065369422
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	16
Number of Questions to be attempted :	16
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0

Official Answer Key: 6406533361687

MAD1_290 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

Consider the following HTML code and select the correct rendered output from the following.

```
<!DOCTYPE html>
<html lang=    >
<head>
    <title>Document</title>
</head>
<body>
    <table border="2">
        <tr>
          <th>One</th>
          <td>Four</td>
        </tr>
        <tr>
          <th> Two</th>
          <td> Three</td>
        </tr>
    </table>
</body>
</html>
```

Options :

(A)

One	Four
Two	Three

(B)

One	**Two**
Four	Three

(C)

One	**Two**
Three	Four

(D)

One	**Four**
Two	Three

Official Answer Key: 6406533362456

MAD1_298 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

Consider the below server response

HTTP/2.0 500 Internal Server Error

Which of the following is/are **True**?

Options :

(F) HTTP/2.0 is the protocol and version respectively

(7) 500 is the port number

(8) 'Internal Server Error" is the status message

(9) 500 is the status code

(10) HTTP/2.0 is the server failure

Question Number : 113 Question Id : 640653996109 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Official Answer Key: 6406533362503, 6406533362504

MAD1_299 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

Consider the following code snippet.

```
@app.route('/name/<type>')
def        .       (type):
    # CODE BLOCK HERE
```

Assume the database has a "product" table represented by class "Product" which has a TEXT column . If we want the server to return a 404 status code when a user goes to the route pe>' with a "type" that does not exist in the database, which of the following lines would give us the desired output?

Options :

(F)

Official Answer Key: 6406533362510, 6406533362511, 6406533362512

MAD1_300 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

Suppose the value of **Method** in form.html is GET (form method is set to GET) and in the input box we write "Application", after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

Options :

(A) http://127.0.0.1:5000/?name=Application

(B) http://127.0.0.1:5000/name=Application

(C) http://127.0.0.1:5000/?name

Official Answer Key: 6406533362464

MAD1_301 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

Suppose the value of **Method** in form.html is POST (form method is set to POST) and in the input box we write "Application", after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

Options :

(B) http://127.0.0.1:5000/?name:Application

(C) http://127.0.0.1:5000/?Application

(D) http://127.0.0.1:5000/?name

(E) http://127.0.0.1:5000/

Java

(F)	64065369437
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	16
Number of Questions to be attempted :	16
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0

Official Answer Key: 6406533362471

• 2.3 CSS Selectors, Specificity & Cascade Rules (5 Questions)

MAD1_036 | Type: MCQ | Marks: +4.5

Source Exam(s): Quiz 1 Jan23 .pdf

Read the statements given below carefully and select the correct option.

Statement 1: If an element that belongs to two different classes is styled externally using both the classes, then for the same attribute, it will acquire styling from the latest class in order.
Statement 2: If an element having an ID and a class is styled externally using both its ID and the class, then for the same attribute, it will acquire styling from the latest selector in order.

Options :

(A) Both statements 1 and 2 are correct

(B) Both statements 1 and 2 are incorrect

(C) Statement 1 is correct but statement 2 is incorrect

(D) Statement 2 is correct but statement 1 is incorrect

Official Answer Key: 6406531562004

MAD1_048 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

Which command would you use to get **detailed information** about the HTTP request and response, **including headers**, and to **help debug the HTTP request** when using curl?

Options :

(B) `curl -i <URL>`

(C) `curl -v <URL>`

(D) `curl --headers <URL>`

(E) `curl -X GET <URL>`

Question Number : 69 Question Id : 6406531113761 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Read the statements given below carefully and select the correct option.

Statement 1: If an element that belongs to two different classes is styled externally using both the classes, then for the same attribute, it will acquire styling from the latest class in order.
Statement 2: If an element having an ID and a class is styled externally using both its ID and the class, then for the same attribute, it will acquire styling from the ID selector.

Options :

(F) Both statements 1 and 2 are correct

(7) Both statements 1 and 2 are incorrect

(8) Statement 1 is correct but statement 2 is incorrect

(9) Statement 2 is correct but statement 1 is incorrect

Official Answer Key: 6406533774148

MAD1_063 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Which command would you use to get **detailed information** about the HTTP request and response, **including headers**, and to **help debug the HTTP request** when using curl?

Options :

(B) `curl -i <URL>`

(C) `curl -v <URL>`

(D) `curl --headers <URL>`

(E) `curl -X GET <URL>`

Question Number : 69 Question Id : 6406531114388 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Read the statements given below carefully and select the correct option.

Statement 1: If an element that belongs to two different classes is styled externally using both the classes, then for the same attribute, it will acquire styling from the latest class in order.
Statement 2: If an element having an ID and a class is styled externally using both its ID and the class, then for the same attribute, it will acquire styling from the ID selector.

Options :

(F) Both statements 1 and 2 are correct

(7) Both statements 1 and 2 are incorrect

(8) Statement 1 is correct but statement 2 is incorrect

(9) Statement 2 is correct but statement 1 is incorrect

Official Answer Key: 6406533775961

MAD1_077 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

Which command would you use to get **detailed information** about the HTTP request and response, **including headers**, and to **help debug the HTTP request** when using curl?

Options :

(B) `curl -i <URL>`

(C) `curl -v <URL>`

(D) `curl --headers <URL>`

(E) `curl -X GET <URL>`

Question Number : 69 Question Id : 6406531114101 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Read the statements given below carefully and select the correct option.

Statement 1: If an element that belongs to two different classes is styled externally using both the classes, then for the same attribute, it will acquire styling from the latest class in order.
Statement 2: If an element having an ID and a class is styled externally using both its ID and the class, then for the same attribute, it will acquire styling from the ID selector.

Options :

(F) Both statements 1 and 2 are correct

(7) Both statements 1 and 2 are incorrect

(8) Statement 1 is correct but statement 2 is incorrect

(9) Statement 2 is correct but statement 1 is incorrect

Official Answer Key: 6406533775132

MAD1_131 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

Consider the following HTML document.

```
<!DOCTYPE html>
<html lang=    >
<head>
    <title>Span</title>
    <style>
        span{
            color: blue;
            Background-color: lime;
        }
    </style>
</head>
<body>
    <span id=     pan">MAD-I is a Diploma level course.</span>
    <span>SE is a Degree level course.</span>
</body>
</html>
```

\..nat will be "he re dered output if an additional style is added via the ID selector given below?

```
#my_span{
         width: 1200px;
         Background-color: lightpink;
        }
```

Options :

(A) MAD-I is a Diploma level course. SE is a Degree level course.

(B) MAD-I is a Diploma level course. SE is a Degree level course.

(C) MAD-I is a Diploma level course. SE is a Degree level course.

(D) MAD-I is a Diploma level course.
SE is a Degree level course.

Official Answer Key: 6406532774752

• 2.4 CSS Box Model & Layouts (Flexbox, Positioning & Spacing) (16 Questions)

MAD1_025 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

Consider the following HTML document with an embedded style sheet.

```
<!DOCTYPE html>
<html>
    <head>
        <title>Quiz 1</title>
        <style type="text/css">
            div{
                padding-left: 10px;
                margin-right: 15px;
                border-style: solid;
                border-width: 5px;
                width: 2000px;
                height: 20px;
                }
        </style>
    </head>
    <body>
        <div>My first Div element</div>
    </body>
</html>
```

Which of the following figures correctly represents the box model of the above HTML document?

Options :

(C)

Official Answer Key: 6406531148376

MAD1_037 | Type: MCQ | Marks: +4.5

Source Exam(s): Quiz 1 Jan23 .pdf, Quiz1 May23 .pdf

Consider the following HTML document.

```
<!DOCTYPE html>
<html lang=    >
<head>
    <style>
        [
            border-style: dashed;
            background-color: orange;
        }
        .boxes{
            border-style: solid;
            border-color: blue;
        }
        div{
            width: 200px;
            border: 4px dotted yellow;
        }
    </style>
</head>
<body>
    <div id='   ox" class="boxes" style="background-color: limegreen;">
        <h1>Hello MAD-I</h1>
    </div>
</body>
</html>
```

How will the browser render the above HTML document?

Options :

(E)

Official Answer Key: 6406531562012

MAD1_049 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

Which of the following options represents a one-to-one relationship in the Entity-Relationship Diagram?

Options :

(A)

(B)

(C)

(D)

(E)

Sub-Section Number :	4
Question Shuffling Allowed :	640653168510
	Yes

Question Number : 71 Question Id : 6406531113763 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the following HTML document:

```
<!DOCTYPE html>
<html lang=    >
<head>
    <style>
            r {
            border-style: dashed;
            background-color: red;
        }
        .content {
            border-style: solid;
            border-color: green;
            background-color: orange;
        }
        div {
            width: 250px;
            border: 3px dotted black;
            border-color: yellow;
        }
    </style>
</head>
<body>
```

Official Answer Key: 6406533774156

MAD1_064 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Which of the following options represents a one-to-one relationship in the Entity-Relationship Diagram?

Options :

(A)

(B)

(C)

(D)

Sub-Section Number :	4
(E)	640653168714
Question Shuffling Allowed :	Yes

Question Number : 71 Question Id : 6406531114390 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the following HTML document:

```
<!DOCTYPE html>
<html lang=    >
<head>
    <style>
            r {
            border-style: dashed;
            background-color: red;
        }
        .content {
            border-style: solid;
            border-color: green;
            background-color: orange;
        }
        div {
            width: 250px;
            border: 3px dotted black;
            border-color: yellow;
        }
    </style>
</head>
<body>
```

Official Answer Key: 6406533775969

MAD1_078 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

Which of the following options represents a one-to-one relationship in the Entity-Relationship Diagram?

Options :

(A)

(B)

(C)

(D)

(E)

Sub-Section Number :	4
Question Shuffling Allowed :	640653168615
	Yes

Question Number : 71 Question Id : 6406531114103 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the following HTML document:

```
<!DOCTYPE html>
<html lang=    >
<head>
    <style>
            r {
            border-style: dashed;
            background-color: red;
        }
        .content {
            border-style: solid;
            border-color: green;
            background-color: orange;
        }
        div {
            width: 250px;
            border: 3px dotted black;
            border-color: yellow;
        }
    </style>
</head>
<body>
```

Official Answer Key: 6406533775140

MAD1_090 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz1 May23 .pdf

Consider a square matrix A given below.

$$A = \begin{bmatrix} 12 & 21 & 25 \\ 11 & 8 & 30 \\ 9 & 19 & 17 \end{bmatrix}$$

Matrix B is another matrix whose elements represent the LSBs of binary equivalent of corresponding elements of matrix A. The determinant of matrix B would be _____.

Options :

(A) 0

(B) 1

(C) 2

(D) 3

Official Answer Key: 6406531929693

MAD1_149 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Consider the following HTML and CSS code:

```
div {
  width: 0px;
  margin: 0px;
}
 {
  width: 200px;
  padding: 15px;
  border: 5px solid black;
}
 {
  margin: 10px;
}
<
<
 :"container" id="box">Hello World</div>
<
```

What is the **total width (approximately)** of the `div` with `class="container"` and `="␣ "`, including **content, padding, and border** but **excluding margin**?

Options :

(A) 200 px

(B) 230 px

(C) 240 px

(D) 250 px

Official Answer Key: 6406535115578

MAD1_162 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Consider the following HTML and CSS code:

```
div {
  width: 0px;
  margin: 0px;
}
 {
  width: 200px;
  padding: 15px;
  border: 5px solid black;
}
 {
  margin: 10px;
}
<
<
 :"container" id="box">Hello World</div>
<
```

What is the **total width (approximately)** of the `div` with `class="container"` and `="␣ "`, including **content, padding, and border** but **excluding margin**?

Options :

(A) 200 px

(B) 230 px

(C) 240 px

(D) 250 px

Official Answer Key: 6406535116421

MAD1_173 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Consider the following HTML and CSS code:

```
div {
  width: 0px;
  margin: 0px;
}
 {
  width: 200px;
  padding: 15px;
  border: 5px solid black;
}
 {
  margin: 10px;
}
<
<
 :"container" id="box">Hello World</div>
<
```

What is the **total width (approximately)** of the `div` with `class="container"` and `="␣ "`, including **content, padding, and border** but **excluding margin**?

Options :

(A) 200 px

(B) 230 px

(C) 240 px

(D) 250 px

Official Answer Key: 6406535118352

MAD1_185 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Consider the following HTML and CSS code:

```
div {
  width: 0px;
  margin: 0px;
}
 {
  width: 200px;
  padding: 15px;
  border: 5px solid black;
}
 {
  margin: 10px;
}
<
<
 :"container" id="box">Hello World</div>
<
```

What is the **total width (approximately)** of the `div` with `class="container"` and `="␣ "`, including **content, padding, and border** but **excluding margin**?

Options :

(A) 200 px

(B) 230 px

(C) 240 px

(D) 250 px

Official Answer Key: 6406535130972

MAD1_197 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Consider the following HTML and CSS code:

```
div {
  width: 0px;
  margin: 0px;
}
 {
  width: 200px;
  padding: 15px;
  border: 5px solid black;
}
 {
  margin: 10px;
}
<
<
 :"container" id="box">Hello World</div>
<
```

What is the **total width (approximately)** of the `div` with `class="container"` and `="␣ "`, including **content, padding, and border** but **excluding margin**?

Options :

(A) 200 px

(B) 230 px

(C) 240 px

(D) 250 px

Official Answer Key: 6406535131868

MAD1_208 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Consider the following HTML and CSS code:

```
div {
  width: 0px;
  margin: 0px;
}
 {
  width: 200px;
  padding: 15px;
  border: 5px solid black;
}
 {
  margin: 10px;
}
<
<
 :"container" id="box">Hello World</div>
<
```

What is the **total width (approximately)** of the `div` with `class="container"` and `="␣ "`, including **content, padding, and border** but **excluding margin**?

Options :

(A) 200 px

(B) 230 px

(C) 240 px

(D) 250 px

Official Answer Key: 6406535132764

MAD1_219 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Consider the following HTML and CSS code:

```
div {
  width: 0px;
  margin: 0px;
}
 {
  width: 200px;
  padding: 15px;
  border: 5px solid black;
}
 {
  margin: 10px;
}
<
<
 :"container" id="box">Hello World</div>
<
```

What is the **total width (approximately)** of the `div` with `class="container"` and `="␣ "`, including **content, padding, and border** but **excluding margin**?

Options :

(A) 200 px

(B) 230 px

(C) 240 px

(D) 250 px

Official Answer Key: 6406535125845

MAD1_231 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Consider the following HTML and CSS code:

```
div {
  width: 0px;
  margin: 0px;
}
 {
  width: 200px;
  padding: 15px;
  border: 5px solid black;
}
 {
  margin: 10px;
}
<
<
 :"container" id="box">Hello World</div>
<
```

What is the **total width (approximately)** of the `div` with `class="container"` and `="␣ "`, including **content, padding, and border** but **excluding margin**?

Options :

(A) 200 px

(B) 230 px

(C) 240 px

(D) 250 px

Official Answer Key: 6406535128337

MAD1_242 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Consider the following HTML and CSS code:

```
div {
  width: 0px;
  margin: 0px;
}
 {
  width: 200px;
  padding: 15px;
  border: 5px solid black;
}
 {
  margin: 10px;
}
<
<
 :"container" id="box">Hello World</div>
<
```

What is the **total width (approximately)** of the `div` with `class="container"` and `="␣ "`, including **content, padding, and border** but **excluding margin**?

Options :

(A) 200 px

(B) 230 px

(C) 240 px

(D) 250 px

Official Answer Key: 6406535129907

MAD1_247 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

What does the term “responsive” mean in the context of web design?

Options :

(B) The ability of a webpage to give visual feedback on any kind of user input.

(C) The ability of a webpage to change its color scheme to “night mode”.

(D) The ability of a webpage to alter its layout and appearance to suit different olutions, etc.

(E) All of these.

Question Number : 103 Question Id : 640653668543 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructior Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Match the following HTTP methods in Column A with their conventional usage in Column B.

Official Answer Key: 6406532239809

WEEK 3: BACKEND PYTHON, FLASK ESSENTIALS, ROUTING & REQUEST HANDLING

• 3.1 Python Web Fundamentals & Command-Line Arguments (29 Questions)

MAD1_009 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

Consider the following Python code snippet:

main.py

```
import sys
def       (v):
   # Missing code 1
v = sys.argv
# Missing code 2
print(output)
```

We are executing above main.py in command line with command `python main.py 1 2 3 4 5` . Which of the following codes when replaced at #Missing code 1 and #Missing code 2 will render the output below:

`1,2,3,4,5`

Options :

(A)

#Missing code 1
```
st = ",".join(v)
return st
```

#Missing code 2
```
output = process(v)
```

(B)

#Missing code 1
```
st = str(join(v),",")
return st
```

#Missing code 2
```
output = process(v)
```

(C)

#Missing code 1
```
st = str(join(v),",")
return st
```

#Missing code 2
```
output = process(v[1:])
```

Official Answer Key: 6406534366214

MAD1_013 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

What will be the output for command line input,

`python webapp.py react type` ?

Options :

(F) `react is a client-side javascript framework`

(7) `react is a server-side python framework`

(8) `Invalid second argument or language not found!`

(9) `Invalid command`

Official Answer Key: 6406534366222

MAD1_014 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

What will be the output for command line input,

`python webapp.py react type` ?

Options :

(F) `react is a client-side javascript framework`

(7) `react is a server-side python framework`

(8) `Invalid second argument or language not found!`

(9) `Invalid command`

Official Answer Key: 6406534366228

MAD1_034 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz 1 Jan23 .pdf

How to represent the number 111_{10} in the binary format?

Options :

(A) 1110000

(B) 1101111

(C) 1100000

(D) 1111000

Official Answer Key: 6406531561992

MAD1_052 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

Consider the following Python code snippet:

Python code: "r ain.py"

```
import sys
movie_ratings = {
        : [5, 1, 2, 1, 1],
        : [7, 3, 1, 2, 1],
      : [4, 3, 2, 2, 1],
}

def              (movie_name):
   avg_ratings = {}
   for movie in movie_ratings:
       if movie == movie_name:
           avg_ratings[movie] = sum(movie_ratings[movie]) /
len(movie_ratings[movie])
           return avg_ratings
   return       not found"

args = sys.argv
print(do_something(args[1]))
```

The above file will result in the output as **"2.8"**, for which of the following command line input?

Options :

(A) ɔython main.py Kalki

(B) ɔython main.py Leo

(C) ɔython main.py Pushpa2

(D) ɔython main.py Jailer

Official Answer Key: 6406533774171

MAD1_053 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

Consider two python files, mad1.py and mad2.py with following code snippets.

File1: mad1.py

```
import sys
import mad2
print(          + ' ' + sys.argv[1])
```

File2: mad2.py

```
import sys
print(          + ' ' + sys.argv[0])
```

What is the output of the following command "python mad1.py mad2.py MAD-I MAD-II"?

Options :

(A)
Welcome MAD-I.py
Welcome MAD-II.py

(B)
Welcome mad2.py
Welcome mad1.py

(C)
Welcome mad1.py
Welcome mad2.py

(D)
Welcome MAD-II.py
Welcome MAD-I.py

Sub-Section Number : 5

(E) 640653168511

Question Shuffling Allowed : Yes

Official Answer Key: 6406533774174

MAD1_066 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Consider the following Python code snippet:

Python code: "r ain.py"

```
import sys
movie_ratings = {
          : [5, 1, 2, 1, 1],
          : [7, 3, 1, 2, 1],
        : [4, 3, 2, 2, 1],
}

def              (movie_name):
   avg_ratings = {}
   for movie in movie_ratings:
       if movie == movie_name:
           avg_ratings[movie] = sum(movie_ratings[movie]) /
len(movie_ratings[movie])
           return avg_ratings
   return        not found"

args = sys.argv
print(do_something(args[1]))
```

The above file will result in the output as **"2.8"**, for which of the following command line input?

Options :

(A) ɔython main.py Kalki

(B) ɔython main.py Leo

(C) ɔython main.py Pushpa2

(D) ɔython main.py Jailer

Official Answer Key: 6406533775984

MAD1_067 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Consider two python files, mad1.py and mad2.py with following code snippets.

File1: mad1.py

```
import sys
import mad2
print(          + ' ' + sys.argv[1])
```

File2: mad2.py

```
import sys
print(          + ' ' + sys.argv[0])
```

What is the output of the following command "python mad1.py mad2.py MAD-I MAD-II"?

Options :

(A)
Welcome MAD-I.py
Welcome MAD-II.py

(B)
Welcome mad2.py
Welcome mad1.py

(C)
Welcome mad1.py
Welcome mad2.py

(D)
Welcome MAD-II.py
Welcome MAD-I.py

Sub-Section Number : 5

(E) 640653168715

Question Shuffling Allowed : Yes

Official Answer Key: 6406533775987

MAD1_080 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

Consider the following Python code snippet:

Python code: "r ain.py"

```
import sys
movie_ratings = {
          : [5, 1, 2, 1, 1],
          : [7, 3, 1, 2, 1],
        : [4, 3, 2, 2, 1],
}

def              (movie_name):
   avg_ratings = {}
   for movie in movie_ratings:
       if movie == movie_name:
           avg_ratings[movie] = sum(movie_ratings[movie]) /
len(movie_ratings[movie])
           return avg_ratings
   return        not found"

args = sys.argv
print(do_something(args[1]))
```

The above file will result in the output as **"2.8"**, for which of the following command line input?

Options :

(A) ɔython main.py Kalki

(B) ɔython main.py Leo

(C) ɔython main.py Pushpa2

(D) ɔython main.py Jailer

Official Answer Key: 6406533775155

MAD1_081 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

Consider two python files, mad1.py and mad2.py with following code snippets.

File1: mad1.py

```
import sys
import mad2
print(          + ' ' + sys.argv[1])
```

File2: mad2.py

```
import sys
print(          + ' ' + sys.argv[0])
```

What is the output of the following command "python mad1.py mad2.py MAD-I MAD-II"?

Options :

(A)
Welcome MAD-I.py
Welcome MAD-II.py

(B)
Welcome mad2.py
Welcome mad1.py

(C)
Welcome mad1.py
Welcome mad2.py

(D)
Welcome MAD-II.py
Welcome MAD-I.py

Sub-Section Number : 5

(E) 640653168616

Question Shuffling Allowed : Yes

Official Answer Key: 6406533775158

MAD1_102 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

Consider the following Python code snippet:

main.py

```
import sys
def       (v):
   # Missing code 1
v = sys.argv
# Missing code 2
print(output)
```

We are executing above main.py in command line with command `python main.py 1 2 3 4 5` . Which of the following codes when replaced at #Missing code 1 and #Missing code 2 will render the output below:

`1,2,3,4,5`

Options :

(A)

```
#Missing code 1
  st = ",".join(v)
  return st

#Missing code 2
output = process(v)
```

(B)

```
#Missing code 1
  st = str(join(v),",")
  return st

#Missing code 2
output = process(v)
```

(C)

```
#Missing code 1
  st = str(join(v),",")
  return st

#Missing code 2
output = process(v[1:])
```

Official Answer Key: 6406534364672

MAD1_105 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

What will be the output for command line input,

`python webapp.py react type` ?

Options :

(F) `react is a client-side javascript framework`

(7) `react is a server-side python framework`

(8) `Invalid second argument or language not found!`

(9) `Invalid command`

Official Answer Key: 6406534364680

MAD1_106 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

What will be the output for command line input,

`python webapp.py react type` ?

Options :

(F) `react is a client-side javascript framework`

(7) `react is a server-side python framework`

(8) `Invalid second argument or language not found!`

(9) `Invalid command`

Official Answer Key: 6406534364686

MAD1_117 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

Consider the following Python code snippet:

main.py

```
import sys
def       (v):
   # Missing code 1
v = sys.argv
# Missing code 2
print(output)
```

We are executing above main.py in command line with command `python main.py 1 2 3 4 5` .
Which of the following codes when replaced at #Missing code 1 and #Missing code 2 will render the output below:

`1,2,3,4,5`

Options :

(A)

#Missing code 1
```
st = ",".join(v)
return st
```
#Missing code 2
```
output = process(v)
```

(B)

#Missing code 1
```
st = str(join(v),",")
return st
```
#Missing code 2
```
output = process(v)
```

(C)

#Missing code 1
```
st = str(join(v),",")
return st
```
#Missing code 2
```
output = process(v[1:])
```

Official Answer Key: 6406534365443

MAD1_120 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

What will be the output for command line input,

`python webapp.py react type` ?

Options :

(F) `react is a client-side javascript framework`

(7) `react is a server-side python framework`

(8) `Invalid second argument or language not found!`

(9) `Invalid command`

Official Answer Key: 6406534365451

MAD1_121 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

What will be the output for command line input,

`python webapp.py react type` ?

Options :

(F) `react is a client-side javascript framework`

(7) `react is a server-side python framework`

(8) `Invalid second argument or language not found!`

(9) `Invalid command`

Official Answer Key: 6406534365457

MAD1_132 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf

Consider the following HTML document.

```
<!DOCTYPE html>
<html lang=    >
<head>
    <title>Span</title>
    <style>
        span{
            color: blue;
            Background-color: lime;
        }
    </style>
</head>
<body>
    <span id=     pan">MAD-I is a Diploma level course.</span>
    <span>SE is a Degree level course.</span>
</body>
</html>
```

\...ser will be `re re ltered output if an additional style is added via the ID selector given below?

```
#my_span{
        width: 1200px;
        Background-color: lightpink;
        }
```

Options :

(A) MAD-I is a Diploma level course. SE is a Degree level course.

(B) MAD-I is a Diploma level course. SE is a Degree level course.

(C) MAD-I is a Diploma level course. SE is a Degree level course.

(D) MAD-I is a Diploma level course.
SE is a Degree level course.

Question Number : 42 Question Id : 640653825031 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following Python code snippet.

F :ename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def    _    ():
    if request.method == 'GET':
        retu m f"Hello from {sys.argv[1]} method"

    elif re uest.method == 'POST':
        return f"Hello from {sys.argv[2]} method"

    else:
        return "Please enter a valid HTTP method"

app.run(debu _ = True)
```

If `he above *task .pp is run using the command `python app.py POST GET` in one terr...nal, `r` at ...d be the output on another terminal for command; `curl -X POST http://127.0.0.1:5000/home` ?

Options :

(A)
```
Hello from POST method
```

(B)
```
Hello from GET method
```

(C)
```
Please enter a valid HTTP method
```

(D)
```
Method not Allowed
```

Official Answer Key: 6406532774772

MAD1_133 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

Consider the following Python code snippet.
F ˜sname: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def _ _ ():
    if request.method == 'GET':
        retu rn f"Hello from {sys.argv[1]} method"

    elif re uest.method == 'POST':
        return f"Hello from {sys.argv[2]} method"

    else:
        return "Please enter a valid HTTP method"

app.run(debu ; = True)
```

If ˜he above ˜lask ..pp is run using the command `python app.py POST GET` in one terr...nal, v ˜ at w..l be the output on another terminal for command;
`curl -X POST http://127.0.0.1:5000/home` ?

Options :

(A) `Hello from POST method`

(B) `Hello from GET method`

(C) `Please enter a valid HTTP method`

(D) `Method not Allowed`

Question Number : 43 Question Id : 640653825032 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Consider the following Python code snippet.

Filena˙ne: nodu◆.py

```
 rom pyhtml import *
import sys
var = sys.˙rgv[1:]

def list_f ˚nc(stack):
    for tech in stack:
        yield span(tech)

doc = html(
        head(title('My Document')),
        bc˚ty(
            h3('Frontend technologies'),
            list_func(var[::-1])
        )
)
print˚ loc.render())
```

How will the b ˛wser render the output of the above code if it is run on the terminal ı ing the command `python module.py HTML CSS JS VUE`?

Official Answer Key: 6406532774776

MAD1_139 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

Consider the following HTML document.

```
<!DOCTYPE html>
<html lang=    >
<head>
    <title>Span</title>
    <style>
        span{
            color: blue;
            Background-color: lime;
        }
    </style>
</head>
<body>
    <span id=     pan">MAD-I is a Diploma level course.</span>
    <span>SE is a Degree level course.</span>
</body>
</html>
```

\...tat will be `he re ltered output if an additional style is added via the ID selector given below?

```
#my_span{
        width: 1200px;
        Background-color: lightpink;
        }
```

Options :

(A) MAD-I is a Diploma level course. SE is a Degree level course.

(B) MAD-I is a Diploma level course. SE is a Degree level course.

(C) MAD-I is a Diploma level course. SE is a Degree level course.

(D) MAD-I is a Diploma level course.
E is a Degree level course.

Question Number : 42 Question Id : 640653827500 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following Python code snippet.

F iname: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def    _    ():
    if request.method == 'GET':
        retu rn f"Hello from {sys.argv[1]} method"

    elif re uest.method == 'POST':
        return f"Hello from {sys.argv[2]} method"

    else:
        return "Please enter a valid HTTP method"

app.run(debu _ = True)
```

If `he above *task .pp is run using the command `python app.py POST GET` in one terr...nal, `r at \...d` be the output on another terminal for command; `curl -X POST http://127.0.0.1:5000/home` ?

Options :

(A)
```
Hello from POST method
```

(B)
```
Hello from GET method
```

(C)
```
Please enter a valid HTTP method
```

(D)
```
Method not Allowed
```

Official Answer Key: 6406532781296

MAD1_141 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 7 July 2024.pdf

Consider the following HTML document.

```
<!DOCTYPE html>
<html lang=    >
<head>
    <title>Span</title>
    <style>
        span{
            color: blue;
            Background-color: lime;
        }
    </style>
</head>
<body>
    <span id=     pan">MAD-I is a Diploma level course.</span>
    <span>SE is a Degree level course.</span>
</body>
</html>
```

\...ser will be `re re ltered output if an additional style is added via the ID selector given below?

```
#my_span{
        width: 1200px;
        Background-color: lightpink;
        }
```

Options :

(A) MAD-I is a Diploma level course. SE is a Degree level course.

(B) MAD-I is a Diploma level course. SE is a Degree level course.

(C) MAD-I is a Diploma level course. SE is a Degree level course.

(D) MAD-I is a Diploma level course.
SE is a Degree level course.

Question Number : 42 Question Id : 64065382778 8 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question
Consider the following Python code snippet.
F :ename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def    _    ():
    if request.method == 'GET':
        retu rn f"Hello from {sys.argv[1]} method"

    elif re uest.method == 'POST':
        return f"Hello from {sys.argv[2]} method"

    else:
        return "Please enter a valid HTTP method"

app.run(debu _ = True)
```

If `he above *task _pp is run using the command `python app.py POST GET` in one terr...nal, `r` an `...d` be the output on another terminal for command; `curl -X POST http://127.0.0.1:5000/home` ?

Options :

(A)
```
Hello from POST method
```

(B)
```
Hello from GET method
```

(C)
```
Please enter a valid HTTP method
```

(D)
```
Method not Allowed
```

Official Answer Key: 6406532782092

MAD1_145 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "**python main.py data.txt output.txt**" when run on terminal?

Options :

(B)

```
Module: utils.py
Processing: data.txt
```

(C)

```
Processing: data.txt
Module: utils.py
```

(D)

```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535115570

MAD1_158 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "**python main.py data.txt output.txt**" when run on terminal?

Options :

(B)

```
Module: utils.py
Processing: data.txt
```

(C)

```
Processing: data.txt
Module: utils.py
```

(D)

```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535116413

MAD1_169 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "**python main.py data.txt output.txt**" when run on terminal?

Options :

(B)

```
Module: utils.py
Processing: data.txt
```

(C)

```
Processing: data.txt
Module: utils.py
```

(D)

```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535118344

MAD1_181 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "**python main.py data.txt output.txt**" when run on terminal?

Options :

(D)

```
Module: utils.py
Processing: data.txt
```

(E)

```
Processing: data.txt
Module: utils.py
```

Official Answer Key: 6406535130964

MAD1_193 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "**python main.py data.txt output.txt**" when run on terminal?

Options :

(D)

```
Module: utils.py
Processing: data.txt
```

(E)

```
Processing: data.txt
Module: utils.py
```

Official Answer Key: 6406535131860

MAD1_204 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "**python main.py data.txt output.txt**" when run on terminal?

Options :

(D)

```
Module: utils.py
Processing: data.txt
```

(E)

```
Processing: data.txt
Module: utils.py
```

Official Answer Key: 6406535132756

MAD1_215 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

Options :

(B)

```
Module: utils.py
Processing: data.txt
```

(C)

```
Processing: data.txt
Module: utils.py
```

(D)

```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535125837

MAD1_227 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

Options :

(B)

```
Module: utils.py
Processing: data.txt
```

(C)

```
Processing: data.txt
Module: utils.py
```

(D)

```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535128329

MAD1_238 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

Options :

(B)

```
Module: utils.py
Processing: data.txt
```

(C)

```
Processing: data.txt
Module: utils.py
```

(D)

```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535129899

• 3.2 Flask Application Setup & App Configuration (31 Questions)

MAD1_001 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

What will be the output of the following HTML code?

```
<!DOCTYPE html>
<html>
<head>
    <title>Test List</title>
</head>
<body>
    <dl>
        <dt>Flask</dt>
        <dd>A Python web framework.</dd>
        <dt>SQLAlchemy</dt>
        <dd>A database toolkit for Python.</dd>
    </dl>
</body>
</html>
```

Options :

(A)

Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

(B)

Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

(C)

Flask
- A Python web framework.
SQLAlchemy
- A database toolkit for Python.

(D)

Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

Official Answer Key: 6406534366179

MAD1_015 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

Given the following code, answer the given subquestions:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

template = """
<!D  TYf"  f  l>
<l_  l>
<l  ad>.i.le>Protein RDA</title></head>
<bo'y>
  <h2>Recc   nded Daily Protein Intake</h2>
  <p>Gender: {{ gender }}</p>
  <p>  ight: {{ weight_raw }}</p>
  <p>Rec   nded Intake: <strong>{{ rda }} grams</strong></p>
<,.o>,r?
<,.t_ l>
"""

def    _   |   (weight_raw):
    if weight_raw.endswith('kg'):
        return float(weight_raw[:-2])
    elif weight_raw.endswith('lb'):
        return float(weight_raw[:-2]) * 0.453592
    return float(weight_raw)

def  t   'n_r' (gender, weight_kg):
    return weight_kg * (0.8 if gender.lower() == 'male' else 0.75)

@app.route('/protein', methods=['GET'])
def   t   l_():
    gender     = request.args.get('gender', 'male')
    weight_raw = request.args.get('weight', '60kg')

    weight_kg  = parse_weight(weight_raw)
    rda        = int(round(protein_rda(gender, weight_kg)))
    return render_template_string(template, gender=gender,
weight_raw=weight_raw, rda=rda)

app.run(debug=True)
```

Sub questions

Question Number : 78 Question Id : 6406531298449 Question Type : MCQ
Correct Marks : 4.5

Question Label : Multiple Choice Question

The Flask route and template given in the main data, what will be rendered in the browser when the user visits

`/protein?gender=FeMaLe&weight=154lb`

Options :

(A)

Recommended Daily Protein Intake

Gender: male

Weight: 60kg

Recommended Intake: **48 grams**

(B)

Recommended Daily Protein Intake

Gender: FeMaLe

Weight: 154lb

Recommended Intake: **52 grams**

(C)

Recommended Daily Protein Intake

Gender: FeMaLe

Weight: 154lb

Recommended Intake: **70 grams**

(D)

Recommended Daily Protein Intake

Gender: female

Weight: 154lb

Recommended Intake: **70 grams**

Official Answer Key: 6406534366231

MAD1_016 | Type: MCQ | Marks: +3.0

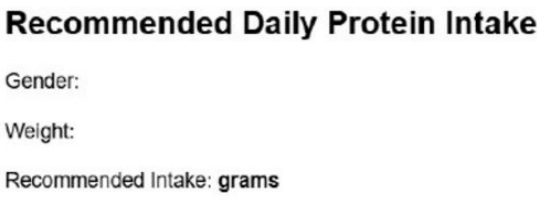
Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

The Flask route and template given in the main data, what will be rendered in the browser when the user visits

```
/protein?gender=FEMALE&weight=60
```

Options :

(A)



(B)

Official Answer Key: 6406534366235

MAD1_041 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024.pdf

Match the URLs in column A with their correct rendered output in Column B with the correct app.py code for the flask application given in the previous question.

Column A	Column B
1. http://127.0.0.1:5000/greet?name=MADI&guest=MADII	a. **Hello, Guest!**
2. http://127.0.0.1:5000/greet?name=MADII&guest=MADI	b. **Hello, MADII!**
3. http://127.0.0.1:5000/greet?	c. **Hello, Guest!**
4. http://127.0.0.1:5000/greet?guest=MADI	d. **Hello, MADI!**

Options :

(B) 1-b, 2-d, 3-a, 4-c

(C) 1-b, 2-a, 3-c, 4-d

(D) 1-d, 2-b, 3-a, 4- c

(E) 1-d, 2-a, 3-c, 4-b

MLF

(F) 64065351357

Section Number : 9

Section type : Online

Mandatory or Optional : Mandatory

Number of Questions : 11

Official Answer Key: 6406532467944

MAD1_043 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024.pdf

Match the URLs in column A with their correct rendered output in Column B with the correct app.py code for the flask application given in the previous question.

Column A	Column B
1. http://127.0.0.1:5000/greet?name=MADI&guest=MADII	a. **Hello, Guest!**
2. http://127.0.0.1:5000/greet?name=MADII&guest=MADI	b. **Hello, MADII!**
3. http://127.0.0.1:5000/greet?	c. **Hello, Guest!**
4. http://127.0.0.1:5000/greet?guest=MADI	d. **Hello, MADI!**

Options :

(B) 1-b, 2-d, 3-a, 4-c

(C) 1-b, 2-a, 3-c, 4-d

(D) 1-d, 2-b, 3-a, 4- c

(E) 1-d, 2-a, 3-c, 4-b

MLF

(F) 64065351372

Section Number : 9

Section type : Online

Mandatory or Optional : Mandatory

Number of Questions : 11

Official Answer Key: 6406532468768

MAD1_045 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024.pdf

Match the URLs in column A with their correct rendered output in Column B with the correct app.py code for the flask application given in the previous question.

Column A	Column B
1. http://127.0.0.1:5000/greet?name=MADI&guest=MADII	a. **Hello, Guest!**
2. http://127.0.0.1:5000/greet?name=MADII&guest=MADI	b. **Hello, MADII!**
3. http://127.0.0.1:5000/greet?	c. **Hello, Guest!**
4. http://127.0.0.1:5000/greet?guest=MADI	d. **Hello, MADI!**

Options :

(B) 1-b, 2-d, 3-a, 4-c

(C) 1-b, 2-a, 3-c, 4-d

(D) 1-d, 2-b, 3-a, 4- c

(E) 1-d, 2-a, 3-c, 4-b

MLF

(F) 64065351387

Section Number : 9

Section type : Online

Mandatory or Optional : Mandatory

Number of Questions : 11

Official Answer Key: 6406532469592

MAD1_094 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

What will be the output of the following HTML code?

```
<!DOCTYPE html>
<html>
<head>
    <title>Test List</title>
</head>
<body>
    <dl>
        <dt>Flask</dt>
        <dd>A Python web framework.</dd>
        <dt>SQLAlchemy</dt>
        <dd>A database toolkit for Python.</dd>
    </dl>
</body>
</html>
```

Options :

(A)
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

(B)
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

(C)
Flask
- A Python web framework.
SQLAlchemy
- A database toolkit for Python.

(D)
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

Official Answer Key: 6406534364637

MAD1_107 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

Given the following code, answer the given subquestions:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

template = """
<!D  TYf"  f  l>
<l_  l>
<l  ad>\.i_le>Protein RDA</title></head>
<bo'/p
  <h2>Recc   nded Daily Protein Intake</h2>
  <p>Gender: {{ gender }}</p>
  <p>  ight: {{ weight_raw }}</p>
  <p>Rec   nded Intake: <strong>{{ rda }} grams</strong></p>
<,.o>,r?
<,.t_ l>
"""

def    _   |   (weight_raw):
    if weight_raw.endswith('kg'):
        return float(weight_raw[:-2])
    elif weight_raw.endswith('lb'):
        return float(weight_raw[:-2]) * 0.453592
    return float(weight_raw)

def  t   'n_r' (gender, weight_kg):
    return weight_kg * (0.8 if gender.lower() == 'male' else 0.75)

@app.route('/protein', methods=['GET'])
def   t   l_():
    gender     = request.args.get('gender', 'male')
    weight_raw = request.args.get('weight', '60kg')

    weight_kg  = parse_weight(weight_raw)
    rda        = int(round(protein_rda(gender, weight_kg)))
    return render_template_string(template, gender=gender,
weight_raw=weight_raw, rda=rda)

app.run(debug=True)
```

Sub questions

Question Number : 78 Question Id : 6406531297909 Question Type : MCQ
Correct Marks : 4.5

Question Label : Multiple Choice Question

The Flask route and template given in the main data, what will be rendered in the browser when the user visits

`/protein?gender=FeMaLe&weight=154lb`

Options :

(A)

Recommended Daily Protein Intake

Gender: male

Weight: 60kg

Recommended Intake: **48 grams**

(B)

Recommended Daily Protein Intake

Gender: FeMaLe

Weight: 154lb

Recommended Intake: **52 grams**

(C)

Recommended Daily Protein Intake

Gender: FeMaLe

Weight: 154lb

Recommended Intake: **70 grams**

(D)

Recommended Daily Protein Intake

Gender: female

Weight: 154lb

Recommended Intake: **70 grams**

Official Answer Key: 6406534364689

MAD1_108 | Type: MCQ | Marks: +3.0

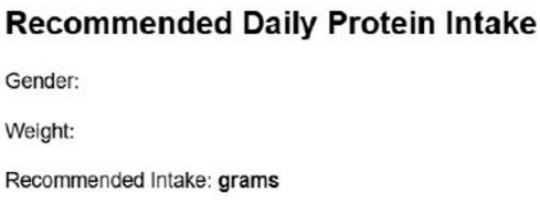
Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

The Flask route and template given in the main data, what will be rendered in the browser when the user visits

```
/protein?gender=FEMALE&weight=60
```

Options :

(A)



(B)

Official Answer Key: 6406534364693

MAD1_109 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

What will be the output of the following HTML code?

```
<!DOCTYPE html>
<html>
<head>
    <title>Test List</title>
</head>
<body>
    <dl>
        <dt>Flask</dt>
        <dd>A Python web framework.</dd>
        <dt>SQLAlchemy</dt>
        <dd>A database toolkit for Python.</dd>
    </dl>
</body>
</html>
```

Options :

(A)
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

(B)
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

(C)
Flask
- A Python web framework.
SQLAlchemy
- A database toolkit for Python.

(D)
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

Official Answer Key: 6406534365408

MAD1_122 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

Given the following code, answer the given subquestions:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

template = """
<!D  TYf"  f  l>
<l_  l>
<l  ad>.i_le>Protein RDA</title></head>
<bo'y>
  <h2>Recc   nded Daily Protein Intake</h2>
  <p>Gender: {{ gender }}</p>
  <p>  ight: {{ weight_raw }}</p>
  <p>Rec   nded Intake: <strong>{{ rda }} grams</strong></p>
<,.o>,r?
<,.t_ l>
"""

def    _   |   (weight_raw):
    if weight_raw.endswith('kg'):
        return float(weight_raw[:-2])
    elif weight_raw.endswith('lb'):
        return float(weight_raw[:-2]) * 0.453592
    return float(weight_raw)

def  t   'n_r' (gender, weight_kg):
    return weight_kg * (0.8 if gender.lower() == 'male' else 0.75)

@app.route('/protein', methods=['GET'])
def   t   l_():
    gender     = request.args.get('gender', 'male')
    weight_raw = request.args.get('weight', '60kg')

    weight_kg  = parse_weight(weight_raw)
    rda        = int(round(protein_rda(gender, weight_kg)))
    return render_template_string(template, gender=gender,
weight_raw=weight_raw, rda=rda)

app.run(debug=True)
```

Sub questions

Question Number : 78 Question Id : 6406531298179 Question Type : MCQ
Correct Marks : 4.5

Question Label : Multiple Choice Question

The Flask route and template given in the main data, what will be rendered in the browser when the user visits

`/protein?gender=FeMaLe&weight=154lb`

Options :

(A)

Recommended Daily Protein Intake

Gender: male

Weight: 60kg

Recommended Intake: **48 grams**

(B)

Recommended Daily Protein Intake

Gender: FeMaLe

Weight: 154lb

Recommended Intake: **52 grams**

(C)

Recommended Daily Protein Intake

Gender: FeMaLe

Weight: 154lb

Recommended Intake: **70 grams**

(D)

Recommended Daily Protein Intake

Gender: female

Weight: 154lb

Recommended Intake: **70 grams**

Official Answer Key: 6406534365460

MAD1_123 | Type: MCQ | Marks: +3.0

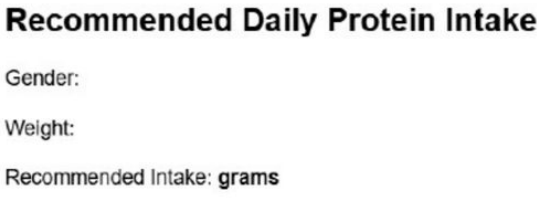
Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

The Flask route and template given in the main data, what will be rendered in the browser when the user visits

```
/protein?gender=FEMALE&weight=60
```

Options :

(A)



(B)

Official Answer Key: 6406534365464

MAD1_142 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Which of the following is the correct syntax to display the total number of courses in a Jinja2 template?

Options :

(A) `{{ courses | len }}`

(B) `{{ courses | length }}`

(C) `{{ len(courses) }}`

(D) None of these

Question Number : 79 Question Id : 6406531527806 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

[redacted]("/[redacted]")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

You are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

Options :

(E) http://127.0.0.1:5000/

(F) http://localhost:0000/

(7) http://0.0.0.0:5000/

(8) http://localhost:5008/

Official Answer Key: 6406535115554

MAD1_153 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Consider the following Python code snippet.

Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

Based on the above data, answer the given subquestions.

Sub questions

Official Answer Key: 6406535115597

MAD1_155 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf, IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

From the given options, select which part of the code (if any) will throw an error first in the terminal and prevent the Flask application from running successfully ?

Options :

(A) `method="GET"` is not correct, it should be `methods=["GET"]`

(B) `def login(id)` can throw error because no `id` is passed in `/login` url

(C) `name` will throw error because no name will be written in the starting of app

(D) There is no error, Flask application will run correctly.

Official Answer Key: 6406535115604

MAD1_156 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Which of the following is the correct syntax to display the total number of courses in a Jinja2 template?

Options :

(A) `{{ courses | len }}`

(B) `{{ courses | length }}`

(C) `{{ len(courses) }}`

(D) None of these

Question Number : 79 Question Id : 6406531528109 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

[redacted]("/[redacted]")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

You are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

Options :

(E) http://127.0.0.1:5000/

(F) http://localhost:0000/

(7) http://0.0.0.0:5000/

(8) http://localhost:5008/

Official Answer Key: 6406535116397

MAD1_166 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Consider the following Python code snippet.

Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

Based on the above data, answer the given subquestions.

Sub questions

Official Answer Key: 6406535116440

MAD1_167 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Which of the following is the correct syntax to display the total number of courses in a Jinja2 template?

Options :

(A) `{{ courses | len }}`

(B) `{{ courses | length }}`

(C) `{{ len(courses) }}`

(D) None of these

Question Number : 79 Question Id : 6406531528771 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

[redacted]("/[redacted]")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

You are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

Options :

(E) http://127.0.0.1:5000/

(F) http://localhost:0000/

(7) http://0.0.0.0:5000/

(8) http://localhost:5008/

Official Answer Key: 6406535118328

MAD1_177 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Consider the following Python code snippet.

Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

Based on the above data, answer the given subquestions.

Sub questions

Official Answer Key: 6406535118371

MAD1_179 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Which of the following is the correct syntax to display the total number of courses in a Jinja2 template?

Options :

(D) `{{ courses | len }}`

(E) `{{ courses | length }}`

(F) `{{ len(courses) }}`

(7) None of these

Question Number : 128 Question Id : 6406531533249 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

[illegible]("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

You are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

Options :

Official Answer Key: 6406535130948

MAD1_189 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Consider the following Python code snippet.

Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

Based on the above data, answer the given subquestions.

Sub questions

Official Answer Key: 6406535130991

MAD1_191 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Which of the following is the correct syntax to display the total number of courses in a Jinja2 template?

Options :

(D) `{{ courses | len }}`

(E) `{{ courses | length }}`

(F) `{{ len(courses) }}`

(7) None of these

Question Number : 128 Question Id : 6406531533574 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

[illegible]("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

You are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

Options :

Official Answer Key: 6406535131844

MAD1_201 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Consider the following Python code snippet.

Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

Based on the above data, answer the given subquestions.

Sub questions

Official Answer Key: 6406535131887

MAD1_202 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Which of the following is the correct syntax to display the total number of courses in a Jinja2 template?

Options :

(D) `{{ courses | len }}`

(E) `{{ courses | length }}`

(F) `{{ len(courses) }}`

(7) None of these

Question Number : 128 Question Id : 6406531533899 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

[redacted]("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

You are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

Options :

Official Answer Key: 6406535132740

MAD1_212 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Consider the following Python code snippet.

Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

Based on the above data, answer the given subquestions.

Sub questions

Official Answer Key: 6406535132783

MAD1_213 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Which of the following is the correct syntax to display the total number of courses in a Jinja2 template?

Options :

(A) `{{ courses | len }}`

(B) `{{ courses | length }}`

(C) `{{ len(courses) }}`

(D) None of these

Question Number : 65 Question Id : 6406531531440 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

[redacted]("/[redacted]")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

You are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

Options :

(E) http://127.0.0.1:5000/

(F) http://localhost:0000/

(7) http://0.0.0.0:5000/

(8) http://localhost:5008/

Official Answer Key: 6406535125821

MAD1_223 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Consider the following Python code snippet.

Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

Based on the above data, answer the given subquestions.

Sub questions

Official Answer Key: 6406535125864

MAD1_225 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Which of the following is the correct syntax to display the total number of courses in a Jinja2 template?

Options :

(A) `{{ courses | len }}`

(B) `{{ courses | length }}`

(C) `{{ len(courses) }}`

(D) None of these

Question Number : 65 Question Id : 6406531532323 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

[redacted]("/[redacted]")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

You are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

Options :

(E) http://127.0.0.1:5000/

(F) http://localhost:0000/

(7) http://0.0.0.0:5000/

(8) http://localhost:5008/

Official Answer Key: 6406535128313

MAD1_235 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Consider the following Python code snippet.

Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

Based on the above data, answer the given subquestions.

Sub questions

Official Answer Key: 6406535128356

MAD1_236 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Which of the following is the correct syntax to display the total number of courses in a Jinja2 template?

Options :

(A) `{{ courses | len }}`

(B) `{{ courses | length }}`

(C) `{{ len(courses) }}`

(D) None of these

Question Number : 65 Question Id : 6406531532886 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

[redacted]("/[redacted]")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

You are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

Options :

(E) http://127.0.0.1:5000/

(F) http://localhost:0000/

(7) http://0.0.0.0:5000/

(8) http://localhost:5008/

Official Answer Key: 6406535129883

MAD1_246 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Consider the following Python code snippet.

Filename: server.py

```
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

Based on the above data, answer the given subquestions.

Sub questions

Official Answer Key: 6406535129926

• 3.3 Routing, Dynamic URL Converters & View Functions (12 Questions)

MAD1_144 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Consider the following statements and select the correct option.

Statement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

Statement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

Options :

(B) Statement 1 is correct, but statement 2 is incorrect.

(C) Statement 2 is correct, but statement 1 is incorrect.

(D) Both statements 1 and 2 are correct

(E) Both statements 1 and 2 are incorrect

Official Answer Key: 6406535115567

MAD1_157 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Consider the following statements and select the correct option.

Statement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

Statement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

Options :

(B) Statement 1 is correct, but statement 2 is incorrect.

(C) Statement 2 is correct, but statement 1 is incorrect.

(D) Both statements 1 and 2 are correct

(E) Both statements 1 and 2 are incorrect

Official Answer Key: 6406535116410

MAD1_168 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Consider the following statements and select the correct option.

Statement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

Statement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

Options :

(B) Statement 1 is correct, but statement 2 is incorrect.

(C) Statement 2 is correct, but statement 1 is incorrect.

(D) Both statements 1 and 2 are correct

(E) Both statements 1 and 2 are incorrect

Official Answer Key: 6406535118341

MAD1_180 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Consider the following statements and select the correct option.

Statement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

Statement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

Options :

(F) Statement 1 is correct, but statement 2 is incorrect.

(7) Statement 2 is correct, but statement 1 is incorrect.

(8) Both statements 1 and 2 are correct

(9) Both statements 1 and 2 are incorrect

Question Number : 130 Question Id : 6406531533252 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following Flask view functions will return a 404 error for the URL: `http://127.0.0.1:5000/profile/user123/active/2024`?

Options :

(10)

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535130961

MAD1_192 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Consider the following statements and select the correct option.

Statement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

Statement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

Options :

(F) Statement 1 is correct, but statement 2 is incorrect.

(7) Statement 2 is correct, but statement 1 is incorrect.

(8) Both statements 1 and 2 are correct

(9) Both statements 1 and 2 are incorrect

Question Number : 130 Question Id : 6406531533577 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following Flask view functions will return a 404 error for the URL: `http://127.0.0.1:5000/profile/user123/active/2024`?

Options :

(10)

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535131857

MAD1_203 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Consider the following statements and select the correct option.

Statement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

Statement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

Options :

(F) Statement 1 is correct, but statement 2 is incorrect.

(7) Statement 2 is correct, but statement 1 is incorrect.

(8) Both statements 1 and 2 are correct

(9) Both statements 1 and 2 are incorrect

Question Number : 130 Question Id : 6406531533902 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following Flask view functions will return a 404 error for the URL: `http://127.0.0.1:5000/profile/user123/active/2024`?

Options :

(10)

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535132753

MAD1_214 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Consider the following statements and select the correct option.

Statement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

Statement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

Options :

(B) Statement 1 is correct, but statement 2 is incorrect.

(C) Statement 2 is correct, but statement 1 is incorrect.

(D) Both statements 1 and 2 are correct

(E) Both statements 1 and 2 are incorrect

Official Answer Key: 6406535125834

MAD1_226 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Consider the following statements and select the correct option.

Statement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

Statement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

Options :

(B) Statement 1 is correct, but statement 2 is incorrect.

(C) Statement 2 is correct, but statement 1 is incorrect.

(D) Both statements 1 and 2 are correct

(E) Both statements 1 and 2 are incorrect

Official Answer Key: 6406535128326

MAD1_237 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Consider the following statements and select the correct option.

Statement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

Statement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

Options :

(B) Statement 1 is correct, but statement 2 is incorrect.

(C) Statement 2 is correct, but statement 1 is incorrect.

(D) Both statements 1 and 2 are correct

(E) Both statements 1 and 2 are incorrect

Official Answer Key: 6406535129896

MAD1_267 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

Which of the following flask view functions will return a 404 error for the URL:

`http://127.0.0.1:5000/communication/landline/0712` ?

Options :

(E)

Official Answer Key: 6406533360948

MAD1_282 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

Which of the following flask view functions will return a 404 error for the URL:

http://127.0.0.1:5000/communication/landline/0712 ?

Options :

(E)

Official Answer Key: 6406533361732

MAD1_295 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

Which of the following flask view functions will return a 404 error for the URL:

http://127.0.0.1:5000/communication/landline/0712 ?

Options :

(E)

Official Answer Key: 6406533362516

• 3.5 Flask Request Object & Data Extraction (3 Questions)

MAD1_269 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

A certain network connection with a bandwidth of 10 Mbps allows making 1000 requests to a server per second. If each request is modified to have an additional data of 8 Kb, what should be the additional bandwidth required to maintain the rate of 2000 requests per second?

Options :

(E) 18 Mbps

(F) 16 Mbps

(7) 26 Mbps

(8) 36 Mbps

Question Number : 110 Question Id : 640653995541 Question Type : MCQ
Correct Marks : 4.5
Question Label : Multiple Choice Question

Consider the following Python code snippet.

Filename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def ():
    if request.method == 'GET':
        return "<h1>Hello from POST</h1>"

    elif request.method == 'POST':
        return "<h1>Hello from GET</h1>"

    else:
        return "<h1>Please enter a valid HTTP method<h1>"

app.run(debug = True)
```

If the above flask app is run using the command `python app.py GET POST` in one terminal, what will be the output on another terminal for command; `curl http://127.0.0.1:5000/home` ?

Options :

(A) `<h1>Hello from GET</h1>`

(B) `<h1>Hello from POST</h1>`

(C) `<h1>Please enter a valid HTTP method</h1>`

(D) `<h1>Method not Allowed</h1>`

Official Answer Key: 6406533360917

MAD1_283 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

A certain network connection with a bandwidth of 10 Mbps allows making 1000 requests to a server per second. If each request is modified to have an additional data of 8 Kb, what should be the additional bandwidth required to maintain the rate of 2000 requests per second?

Options :

(E) 18 Mbps

(F) 16 Mbps

(7) 26 Mbps

(8) 36 Mbps

Question Number : 110 Question Id : 640653995823 Question Type : MCQ
Correct Marks : 4.5
Question Label : Multiple Choice Question

Consider the following Python code snippet.

Filename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def ():
    if request.method == 'GET':
        return "<h1>Hello from POST</h1>"

    elif request.method == 'POST':
        return "<h1>Hello from GET</h1>"

    else:
        return "<h1>Please enter a valid HTTP method<h1>"

app.run(debug = True)
```

If the above flask app is run using the command `python app.py GET POST` in one terminal, what will be the output on another terminal for command; `curl http://127.0.0.1:5000/home` ?

Options :

(A) `<h1>Hello from GET</h1>`

(B) `<h1>Hello from POST</h1>`

(C) `<h1>Please enter a valid HTTP method</h1>`

(D) `<h1>Method not Allowed</h1>`

Official Answer Key: 6406533361701

MAD1_296 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

A certain network connection with a bandwidth of 10 Mbps allows making 1000 requests to a server per second. If each request is modified to have an additional data of 8 Kb, what should be the additional bandwidth required to maintain the rate of 2000 requests per second?

Options :

(E) 18 Mbps

(F) 16 Mbps

(7) 26 Mbps

(8) 36 Mbps

Question Number : 110 Question Id : 640653996105 Question Type : MCQ
Correct Marks : 4.5
Question Label : Multiple Choice Question

Consider the following Python code snippet.

Filename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def ():
    if request.method == 'GET':
        return "<h1>Hello from POST</h1>"

    elif request.method == 'POST':
        return "<h1>Hello from GET</h1>"

    else:
        return "<h1>Please enter a valid HTTP method<h1>"

app.run(debug = True)
```

If the above flask app is run using the command `python app.py GET POST` in one terminal, what will be the output on another terminal for command; `curl http://127.0.0.1:5000/home` ?

Options :

(A) `<h1>Hello from GET</h1>`

(B) `<h1>Hello from POST</h1>`

(C) `<h1>Please enter a valid HTTP method</h1>`

(D) `<h1>Method not Allowed</h1>`

Official Answer Key: 6406533362485

• 3.6 Flask Responses, Redirects & Error Handling (3 Questions)

MAD1_006 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

Consider the following Flask code snippet:

```
from flask import Flask, url_for, request, redirect
app = Flask(__name__)

@app.route("/signin/user")
def       ):
   return url_for("login")

@app.route("/")
def     ():
   return redirect(url_for("login"))

app.run(debug=True)
```

Assume the above code is running on server http://127.0.0.1:5000/ . What will be the output on a web browser for the URL: http://127.0.0.1:5000/ ?

Options :

(A) http://127.0.0.1:5000/signin/user

(B) /signin/user

(C) http://127.0.0.1:5000

(D) Not Found

Official Answer Key: 6406534366200

MAD1_099 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

Consider the following Flask code snippet:

```
from flask import Flask, url_for, request, redirect
app = Flask(__name__)

@app.route("/signin/user")
def       ):
   return url_for("login")

@app.route("/")
def     ():
   return redirect(url_for("login"))

app.run(debug=True)
```

Assume the above code is running on server http://127.0.0.1:5000/ . What will be the output on a web browser for the URL: http://127.0.0.1:5000/ ?

Options :

(A) http://127.0.0.1:5000/signin/user

(B) /signin/user

(C) http://127.0.0.1:5000

(D) Not Found

Official Answer Key: 6406534364658

MAD1_114 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

Consider the following Flask code snippet:

```
from flask import Flask, url_for, request, redirect
app = Flask(__name__)

@app.route("/signin/user")
def       ):
   return url_for("login")

@app.route("/")
def     ():
   return redirect(url_for("login"))

app.run(debug=True)
```

Assume the above code is running on server http://127.0.0.1:5000/ . What will be the output on a web browser for the URL: http://127.0.0.1:5000/ ?

Options :

(A) http://127.0.0.1:5000/signin/user

(B) /signin/user

(C) http://127.0.0.1:5000

(D) Not Found

Official Answer Key: 6406534365429

WEEK 4: JINJA2 TEMPLATING, DYNAMIC RENDERING, SQLITE & CRUD APPLICATIONS

• 4.1 Jinja2 Templating Basics (Variables & Control Flow) (44 Questions)

MAD1_002 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

Mayur and Manoj are playing a friendly cricket match. You are building a Flask web application to display their scores. Which of the following Flask route functions will correctly display the score when the user visits http://127.0.0.1/score ?(Assume that the score.html exists)

Options :

(A)

```
@app.route('/score')
def score():
    return render('score.html', mayur=50, manoj=45)
```

(B)

```
@app.route('/score')
def score():
    return render_template('score.html', mayur=50, manoj=45)
```

(C)

```
def score():
    return render_template('score.html', mayur=50, manoj=45)
@app.route('/score')
```

(D)

```
@app.route('/score')
def score():
    return 'score.html', mayur=50, manoj=45
```

Official Answer Key: 6406534366184

MAD1_004 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

Consider the following Python code snippet.

Filename: code.py

```
from string import Template as string_template
from jinja2 import Template as jinja_template

template1 = string_template('Hello $name, welcome to $city!')
template2 = jinja_template('{{product}} costs {{price}}')

result1 = template1.substitute(name='Alice')
result2 = template1.substitute(name='Alice', city='Boston',
state=          tts')
result3 = template2.render({"product": "laptop"})
result4 = template2.render({"product": "laptop", "price": "45000",
        :     es"})

print(result1)
print(result2)
print(result3)
print(result4)
```

Which of the following statements will throw a key error?

Options :

(A) `template1.substitute(name='Alice')`

(B) `template1.substitute(name='Alice', city='Boston', state='Massachusetts')`

(C) `template2.render({"product": "laptop"})`

(D) `template2.render({"product": "laptop", "price": "45000", "currency": "rupees"})`

Official Answer Key: 6406534366191

MAD1_008 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

Consider the following Python file "output.py"

File: "output.py"

```
from jinja2 import Template
output = :TYPE html>


 g Jinja2 implementation-->


l = [5, 4, 3, 2, 1]
t = Template(output)
output = t.render(sl=l)
print(output)
```

Which of the following code snippet when placed at ***<!--Missing Jinja2 implementation-->*** will render the below unordered list as shown below?

- 1
- 2
- 3
- 4
- 5

Options :

(A)

```
{%for item in sl|sort(reverse=True)%}
    <li>{{item}}</li>
{%endfor%}
```

(B)

```
{%for item in sl%}
    <li>{{item}}</li>
{%endfor%}
```

(C)

Official Answer Key: 6406534366209

MAD1_011 | Type: SA | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

Consider a client which is located 3000 km from the server makes a request through the cable. Suddenly after the request reaches the server, the cable breaks and the response is now to be sent to the client via air. This change of medium caused an additional delay of 47 ms at the server end. How long will the client have to wait before receiving the response (in ms)? (speed on cable = 2e8 m/s and in air 3e8 m/s)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

Sub-Section Number :	6
(A)	640653206742
Question Shuffling Allowed :	No

Question Id : 6406531298444 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (75 to 77)

Question Label : Comprehension

Consider the following Python code snippet and answer the given subquestions

Filename: webapp.py

```
import sys
from jinja2 import Template

args = sys.argv
frameworks = {
    'djia_o': 'python',
    'react': 'javascript',
    'flask': 'python',
    'vue': 'javascript'
}

languages = {
    'python': 'server-side',
    'javascript': 'client-side'
}

template = Template("{{ framework }} is a {{ lang_type }} {{ language }}
framework.")

if len(args) >= 3 and args[1] in frameworks:
    framework_name = args[1]
    language = frameworks[framework_name]

    if args[2] == 'type' and language in languages:
        lang_type = languages[language]
        print(template.render(framework=framework_name, language=language,
lang_type=lang_type))
    else:
        print("Invalid second argument or language not found!")
else:
    print("Invalid command")
```

Sub questions

Official Answer Key: 72

MAD1_051 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

Consider the following HTML code:

```
<!DOCTYPE html>
<html lang=  >
<head>
    <title>Document</title>
    <style>
        div {
            color: blue;
        }
            {
            color: yellow;
        }
        .s2{
            color: green;
        }
        h3{
            color: red;
        }
    </style>
</head>
<body>
    <div class=  2" id="s1"><h3>IIITM Online BS Degree Program</h3></div>
</body>
</html>
```

What is the .olour of the printed text "IITM Online BS Degree Program"?

Options :

(A) 3LUE
(B) /ELLOW
(C) 3REEN
(D) RED

Question Number : 73 Question Id : 6406531113765 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Consider the template code and Python code given below:

Template co ^s: "output.html"

```
<!DOCTYPE html>
<html lang=  >
<head>
</head>
<body>
    <ol>
        {%for p in products%}
            {%if "i" in p.name|string %}
                <li>{{p.id}}, {{p.name}}, {{p.qty}}, {{p.price}}</li>
            {%endif%}
        {%endfor%}
    </ol>
</body>
</html>
```

Python code: "main.py"

```
from jinja2 import Template

products = [
    {  '   :    , "name": "Food", "qty": 78, "price": 100},
```

Official Answer Key: 6406533774165

MAD1_059 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

What will be the output of the given Python code snippet on the terminal for the command `python app.py string` ?

Options :

(A)
```
The book title is Python Basics
and it costs $20.
```

(B)
```
The book title is $title and it
costs $20.
```

(C)
```
The book title is Python Basics
and it costs {{price}}.
```

(D)
```
The book title is $title and it
costs {{price}}.
```

Official Answer Key: 6406533774197

MAD1_060 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

What will be the output of the given Python code snippet on the terminal for the command `python app.py string` ?

Options :

(A)
```
The book title is Python Basics
and it costs $20.
```

(B)
```
The book title is $title and it
costs $20.
```

(C)
```
The book title is Python Basics
and it costs {{price}}.
```

(D)
```
The book title is $title and it
costs {{price}}.
```

Official Answer Key: 6406533774203

MAD1_065 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

Consider the following HTML code:

```
<!DOCTYPE html>
<html lang=  >
<head>
    <title>Document</title>
    <style>
        div {
            color: blue;
        }
            {
            color: yellow;
        }
        .s2{
            color: green;
        }
        h3{
            color: red;
        }
    </style>
</head>
<body>
    <div class=  2" id="s1"><h3>IITM Online BS Degree Program</h3></div>
</body>
</html>
```

What is the .olour of the printed text "IITM Online BS Degree Program"?

Options :

(A) 3LUE
(B) /ELLOW
(C) 3REEN
(D) RED

Question Number : 73 Question Id : 6406531114392 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the template code and Python code given below:

Template co ^s: "output.html"

```
<!DOCTYPE html>
<html lang=  >
<head>
</head>
<body>
    <ol>
        {%for p in products%}
            {%if "i" in p.name|string %}
                <li>{{p.id}}, {{p.name}}, {{p.qty}}, {{p.price}}</li>
            {%endif%}
        {%endfor%}
    </ol>
</body>
</html>
```

Python code: "main.py"

```
from jinja2 import Template

products = [
    {  :  , "name": "Food", "qty": 78, "price": 100},
```

Official Answer Key: 6406533775978

MAD1_073 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

What will be the output of the given Python code snippet on the terminal for the command `python app.py string` ?

Options :

(A)
```
The book title is Python Basics
and it costs $20.
```

(B)
```
The book title is $title and it
costs $20.
```

(C)
```
The book title is Python Basics
and it costs {{price}}.
```

(D)
```
The book title is $title and it
costs {{price}}.
```

Official Answer Key: 6406533776010

MAD1_074 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

What will be the output of the given Python code snippet on the terminal for the command `python app.py string` ?

Options :

(A)
```
The book title is Python Basics
and it costs $20.
```

(B)
```
The book title is $title and it
costs $20.
```

(C)
```
The book title is Python Basics
and it costs {{price}}.
```

(D)
```
The book title is $title and it
costs {{price}}.
```

Official Answer Key: 6406533776016

MAD1_079 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

Consider the following HTML code:

```
<!DOCTYPE html>
<html lang=  >
<head>
    <title>Document</title>
    <style>
        div {
            color: blue;
        }
            {
            color: yellow;
        }
        .s2{
            color: green;
        }
        h3{
            color: red;
        }
    </style>
</head>
<body>
    <div class=  2" id="s1"><h3>IITM Online BS Degree Program</h3></div>
</body>
</html>
```

What is the .olour of the printed text "IITM Online BS Degree Program"?

Options :

(A) 3LUE

(B) /ELLOW

(C) GREEN

(D) RED

Question Number : 73 Question Id : 6406531114105 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Consider the template code and Python code given below:

Template co ^s: "output.html"

```
<!DOCTYPE html>
<html lang=  >
<head>
</head>
<body>
    <ol>
        {%for p in products%}
            {%if "i" in p.name|string %}
                <li>{{p.id}}, {{p.name}}, {{p.qty}}, {{p.price}}</li>
            {%endif%}
        {%endfor%}
    </ol>
</body>
</html>
```

Python code: "main.py"

```
from jinja2 import Template

products = [
    {  '   :      , "name": "Food", "qty": 78, "price": 100},
```

Official Answer Key: 6406533775149

MAD1_087 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

What will be the output of the given Python code snippet on the terminal for the command `python app.py string` ?

Options :

(A)
```
The book title is Python Basics
and it costs $20.
```

(B)
```
The book title is $title and it
costs $20.
```

(C)
```
The book title is Python Basics
and it costs {{price}}.
```

(D)
```
The book title is $title and it
costs {{price}}.
```

Official Answer Key: 6406533775181

MAD1_088 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

What will be the output of the given Python code snippet on the terminal for the command `python app.py string` ?

Options :

(A)
```
The book title is Python Basics
and it costs $20.
```

(B)
```
The book title is $title and it
costs $20.
```

(C)
```
The book title is Python Basics
and it costs {{price}}.
```

(D)
```
The book title is $title and it
costs {{price}}.
```

Official Answer Key: 6406533775187

MAD1_092 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz1 May23 .pdf

A mobile client starts from and is cruising away continuously at 60 kmph from the network tower whose network range is 40 km and bandwidth is 120 Mbps. How much data (in Gigabytes) will be consumed by the client who is continuously using the entire bandwidth before completely moving out of the network?

[Take 1 Byte = 8 bits, 1 KB = 1000 Bytes, 1 MB = 1000 Kilobytes and so on.]

[Consider the speed of light in air to be 3×10^8 m/sec.]

Options :

(A) 24

(B) 28.8

(C) 36

(D) 288

Official Answer Key: 6406531929672

MAD1_095 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

Mayur and Manoj are playing a friendly cricket match. You are building a Flask web application to display their scores. Which of the following Flask route functions will correctly display the score when the user visits http://127.0.0.1/score ?(Assume that the score.html exists)

Options :

(A)

```
@app.route('/score')
def score():
    return render('score.html', mayur=50, manoj=45)
```

(B)

```
@app.route('/score')
def score():
    return render_template('score.html', mayur=50, manoj=45)
```

(C)

```
def score():
    return render_template('score.html', mayur=50, manoj=45)
@app.route('/score')
```

(D)

```
@app.route('/score')
def score():
    return 'score.html', mayur=50, manoj=45
```

Official Answer Key: 6406534364642

MAD1_097 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

Consider the following Python code snippet.

Filename: code.py

```
from string import Template as string_template
from jinja2 import Template as jinja_template

template1 = string_template('Hello $name, welcome to $city!')
template2 = jinja_template('{{product}} costs {{price}}')

result1 = template1.substitute(name='Alice')
result2 = template1.substitute(name='Alice', city='Boston',
state=          :tts')
result3 = template2.render({"product": "laptop"})
result4 = template2.render({"product": "laptop", "price": "45000",
        :     :es"})

print(result1)
print(result2)
print(result3)
print(result4)
```

Which of the following statements will throw a key error?

Options :

(A)
```
template1.substitute(name='Alice')
```

(B)
```
template1.substitute(name='Alice', city='Boston', state='Massachusetts')
```

(C)
```
template2.render({"product": "laptop"})
```

(D)
```
template2.render({"product": "laptop", "price": "45000", "currency": "rupees"})
```

Official Answer Key: 6406534364649

MAD1_101 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

Consider the following Python file "output.py"

File: "output.py"

```
from jinja2 import Template
output =          :TYPE html>


          -       ıg Jinja2 implementation-->


l = [5, 4, 3, 2, 1]
t = Template(output)
output = t.render(sl=l)
print(output)
```

Which of the following code snippet when placed at ***<!--Missing Jinja2 implementation-->*** will render the below unordered list as shown below?

- 1
- 2
- 3
- 4
- 5

Options :

(A)

```
{%for item in sl|sort(reverse=True)%}
    <li>{{item}}</li>
{%endfor%}
```

(B)

```
{%for item in sl%}
    <li>{{item}}</li>
{%endfor%}
```

(C)

Official Answer Key: 6406534364667

MAD1_103 | Type: SA | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

Consider a client which is located 3000 km from the server makes a request through the cable. Suddenly after the request reaches the server, the cable breaks and the response is now to be sent to the client via air. This change of medium caused an additional delay of 47 ms at the server end. How long will the client have to wait before receiving the response (in ms)? (speed on cable = 2e8 m/s and in air 3e8 m/s)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

Sub-Section Number : 6
(A) 640653206540
Question Shuffling Allowed : No

Question Id : 6406531297904 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (75 to 77)

Question Label : Comprehension

Consider the following Python code snippet and answer the given subquestions

Filename: webapp.py

```
import sys
from jinja2 import Template

args = sys.argv
frameworks = {
    'djia_o': 'python',
    'react': 'javascript',
    'flask': 'python',
    'vue': 'javascript'
}

languages = {
    'python': 'server-side',
    'javascript': 'client-side'
}

template = Template("{{ framework }} is a {{ lang_type }} {{ language }}
framework.")

if len(args) >= 3 and args[1] in frameworks:
    framework_name = args[1]
    language = frameworks[framework_name]

    if args[2] == 'type' and language in languages:
        lang_type = languages[language]
        print(template.render(framework=framework_name, language=language,
lang_type=lang_type))
    else:
        print("Invalid second argument or language not found!")
else:
    print("Invalid command")
```

Sub questions

Official Answer Key: 72

MAD1_110 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

Mayur and Manoj are playing a friendly cricket match. You are building a Flask web application to display their scores. Which of the following Flask route functions will correctly display the score when the user visits http://127.0.0.1/score ?(Assume that the score.html exists)

Options :

(A)

```
@app.route('/score')
def score():
    return render('score.html', mayur=50, manoj=45)
```

(B)

```
@app.route('/score')
def score():
    return render_template('score.html', mayur=50, manoj=45)
```

(C)

```
def score():
    return render_template('score.html', mayur=50, manoj=45)
@app.route('/score')
```

(D)

```
@app.route('/score')
def score():
    return 'score.html', mayur=50, manoj=45
```

Official Answer Key: 6406534365413

MAD1_112 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

Consider the following Python code snippet.

Filename: code.py

```
from string import Template as string_template
from jinja2 import Template as jinja_template

template1 = string_template('Hello $name, welcome to $city!')
template2 = jinja_template('{{product}} costs {{price}}')

result1 = template1.substitute(name='Alice')
result2 = template1.substitute(name='Alice', city='Boston',
state=          :tts')
result3 = template2.render({"product": "laptop"})
result4 = template2.render({"product": "laptop", "price": "45000",
        :    :es"})

print(result1)
print(result2)
print(result3)
print(result4)
```

Which of the following statements will throw a key error?

Options :

(A) `template1.substitute(name='Alice')`

(B) `template1.substitute(name='Alice', city='Boston', state='Massachusetts')`

(C) `template2.render({"product": "laptop"})`

(D) `template2.render({"product": "laptop", "price": "45000", "currency": "rupees"})`

Official Answer Key: 6406534365420

MAD1_116 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

Consider the following Python file "output.py"

File: "output.py"

```
from jinja2 import Template
output = TYPE html>

g Jinja2 implementation-->

l = [5, 4, 3, 2, 1]
t = Template(output)
output = t.render(sl=l)
print(output)
```

Which of the following code snippet when placed at **<!--Missing Jinja2 implementation-->** will render the below unordered list as shown below?

- 1
- 2
- 3
- 4
- 5

Options :

(A)

```
{%for item in sl|sort(reverse=True)%}
    <li>{{item}}</li>
{%endfor%}
```

(B)

```
{%for item in sl%}
    <li>{{item}}</li>
{%endfor%}
```

(C)

Official Answer Key: 6406534365438

MAD1_118 | Type: SA | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

Consider a client which is located 3000 km from the server makes a request through the cable. Suddenly after the request reaches the server, the cable breaks and the response is now to be sent to the client via air. This change of medium caused an additional delay of 47 ms at the server end. How long will the client have to wait before receiving the response (in ms)? (speed on cable = 2e8 m/s and in air 3e8 m/s)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

Sub-Section Number : 6
(A) 640653206641
Question Shuffling Allowed : No

Question Id : 6406531298174 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (75 to 77)

Question Label : Comprehension

Consider the following Python code snippet and answer the given subquestions

Filename: webapp.py

```
import sys
from jinja2 import Template

args = sys.argv
frameworks = {
    'djia_o': 'python',
    'react': 'javascript',
    'flask': 'python',
    'vue': 'javascript'
}

languages = {
    'python': 'server-side',
    'javascript': 'client-side'
}

template = Template("{{ framework }} is a {{ lang_type }} {{ language }}
framework.")

if len(args) >= 3 and args[1] in frameworks:
    framework_name = args[1]
    language = frameworks[framework_name]

    if args[2] == 'type' and language in languages:
        lang_type = languages[language]
        print(template.render(framework=framework_name, language=language,
lang_type=lang_type))
    else:
        print("Invalid second argument or language not found!")
else:
    print("Invalid command")
```

Sub questions

Official Answer Key: 72

MAD1_126 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

What will be the output of the following Python code?

```
from jinja2 import Template
data = {     : 101, "prod_name": "Groceries", "sales_amt": 5600,
                : 5}

def     (amount, p):
   return amount * p / 100

a = foo(data[   ales_amt"], data["sales_bonus"])
t = Template(
                ns, you got bonus {{ bonus }} by selling product
                me']}}"
)
output = t.render(data=data, bonus=a)
prir (output)
```

Options :

(E) `Congratulations, you got bonus 70.0 by selling product Groceries`

(F) `Congratulations, you got bonus 140.0 by selling product Groceries`

(7) `Congratulations, you got bonus 280.0 by selling product Groceries`

(8) `Congratulations, you got bonus 560.0 by selling product Groceries`

Official Answer Key: 6406532774761

MAD1_135 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

Consider a client A that makes a request to server B that fetches the information from Datacenter C which in turn fetches other required information from the Datacenter D. The distance between the client A, server B and datacenters C and D within the network and the speed of information in the medium connecting these nodes is mentioned in the table below.

``odes	A to B	B to C	C to D
Distance (m)	12000	8500	7500
Spe ` of Infor ation(m/s)	3×10^8	2×10^8	1.5×10^8

Calculate the round trip latency (in milliseconds) of the network?

Options :

(C) 187

(D) 132.5

(E) 265

(F) 150

Question Number : 46 Question Id : 640653825035 Question Type : MCQ

Correct Marks : 4.5

Question Label : Multiple Choice Question

Consider the following flask app and the bodies of two index.html files shown in the figure below.

app.py

```
from flask import Flask, render_template
import sys

app = ========= app object initialization here ==========

@app.route('/home')
def      ():
    return render_template('index.html')

app.run()
```

app_templates/index.html

```
<h3>Home page</h3>
<p>MAD I focuses on backend development</p>
<h5>IITM ONLINE DEGREE</h5>
```

root_templates/index.html

```
<h3>Home page</h3>
<p>Vue JS is a frontend technology</p>
<h5>IITM BS DEGREE</h5>
```

Official Answer Key: 6406532774790

MAD1_150 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

[illegible].ro[illegible]'/'[illegible]ethods=['GET', 'POST'])
def home([illegible]:
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535115582

MAD1_154 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf, IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

If the given Flask app is run using the command

```
python server.py GET PUT POST
```

in one terminal, what will be the output on another terminal for the command

```
curl -X DELETE http://127.0.0.1:5000/api/data
```
?

Options :

(A) `<h1>Data deleted successfully</h1>`

(B) `<h1>Unsupported operation</h1>`

(C) `<h1>Method not Allowed</h1>`

(D) `<h1>404 Not Found</h1>`

Question Id : 6406531527820 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Official Answer Key: 6406535115602

MAD1_163 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

[illegible].ro[illegible]'/'[illegible]ethods=['GET', 'POST'])
def home([illegible]:
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535116425

MAD1_174 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

[illegible].ro[illegible]'/'[illegible]ethods=['GET', 'POST'])
def home([illegible]:
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535118356

MAD1_178 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf, IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Which of the following is the correct syntax to display the total number of courses in a Jinja2 template?

Options :

(D) `{{ courses | len }}`

(E) `{{ courses | length }}`

(F) `{{ len(courses) }}`

(7) None of these

Question Number : 128 Question Id : 6406531533249 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

[redacted]("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

You are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

Options :

Official Answer Key: 6406535130942

MAD1_186 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

[illegible].ro[illegible]'/'[illegible]ethods=['GET', 'POST'])
def home([illegible]:
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535130976

MAD1_190 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf, IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

If the given Flask app is run using the command

```
python server.py GET PUT POST
```

in one terminal, what will be the output on another terminal for the command

```
curl -X DELETE http://127.0.0.1:5000/api/data
```

?

Options :

(A) `<h1>Data deleted successfully</h1>`

(B) `<h1>Unsupported operation</h1>`

(C) `<h1>Method not Allowed</h1>`

(D) `<h1>404 Not Found</h1>`

Question Id : 6406531533263 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Official Answer Key: 6406535130996

MAD1_198 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

[illegible].ro[illegible]'/'[illegible]ethods=['GET', 'POST'])
def home([illegible]:
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535131872

MAD1_209 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

[illegible].ro[illegible]'/'[illegible]ethods=['GET', 'POST'])
def home([illegible]:
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535132768

MAD1_220 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

[illegible].ro[illegible]'/'[illegible]ethods=['GET', 'POST'])
def home([illegible]:
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535125849

MAD1_224 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf, IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

If the given Flask app is run using the command

```
python server.py GET PUT POST
```

in one terminal, what will be the output on another terminal for the command

```
curl -X DELETE http://127.0.0.1:5000/api/data
```
?

Options :

(A) `<h1>Data deleted successfully</h1>`

(B) `<h1>Unsupported operation</h1>`

(C) `<h1>Method not Allowed</h1>`

(D) `<h1>404 Not Found</h1>`

Question Id : 6406531531454 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Official Answer Key: 6406535125869

MAD1_232 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

[illegible].ro[illegible]'/'[illegible]ethods=['GET', 'POST'])
def home([illegible]:
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535128341

MAD1_243 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

[illegible].ro[illegible]'/'[illegible]ethods=['GET', 'POST'])
def home([illegible]:
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535129911

MAD1_252 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

Consider the following Python code snippet "code.py".

Filename: code.py

```
import sys
from jinja2 import Template

vars = sys.argv
myTech = {   d 1': 'backend', 'mad 2': 'frontend'}

temp = Template("This course focuses on {{var}} development.")

if vars[1] == 'mad 1':
    print(temp.render(myTech['mad 1']))
e if v rs[1] == 'mad 2':
    print(temp.render(myTech['mad 2']))
e se:
    print(   ntion the course name specifically.")
```

What will be the output on the terminal for input: `python code.py mad 1` ?

Options :

(A) `This course focuses on frontend development.`

(B) `This course focuses on backend development.`

(C) `Mention the course name specifically.`

(D) `IndexError: list index out of range`

Official Answer Key: 6406532239869

MAD1_263 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

Consider the following code block and select the correct option regarding the output when we run the python file app.py.

app.py

```
from jinja2 import Template

template=  '
           e to {Degree} DEGREE
           level to clear is {{level}}
           '

rendered=Template(template)
output=rendered.render(Degree="IITM BS",level="Foundation")
print(output)
```

Options :

(A)
Welcome to IITM BS DEGREE
First level to clear is {{ Level }}

(B)
Welcome to {Degree} DEGREE
First level to clear is

(C)
Welcome to IITM BS DEGREE
First level to clear is Foundation

(D)
Welcome to DEGREE
First level to clear is {{ Level }}

Official Answer Key: 6406533360893

MAD1_266 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

Consider the following Python code and select correct output.

```
from jinja2 import Template

class Loan:

   def __ __(self, accno, loan_amount, interest_rate):
       self.accno = accno
       self.loan_amount = loan_amount
       self.interest_rate = interest_rate

   def _ _details(self):
       return (self.accno, self.loan_amount, self.interest_rate)

loan1 = Loan(101, 45900, 5)
loan2 = Loan(102, 170000, 6)

tm1 = Template("Account1 : {{l1.get_loan_details()}}")
tm2 = Template("Account2 : {{l2.get_loan_details()}}")

print(tm1.render(l1=loan1))
print(tm2.render(l2=loan2))
```

Options :

(A) **Account1 : (102, 170000, 6)**
Account2 : (101, 45900, 5)

(B) **Account2 : (102, 170000, 6)**

(C) **Account1 : (101, 45900, 5)**

(D) **Account1 : (101, 45900, 5)**
Account2 : (102, 170000, 6)

Official Answer Key: 6406533360932

MAD1_278 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

Consider the following code block and select the correct option regarding the output when we run the python file app.py.

app.py

```
from jinja2 import Template

template=  '
           e to {Degree} DEGREE
           level to clear is {{level}}
           '

rendered=Template(template)
output=rendered.render(Degree="IITM BS",level="Foundation")
print(output)
```

Options :

(A)
Welcome to IITM BS DEGREE
First level to clear is {{ Level }}

(B)
Welcome to {Degree} DEGREE
First level to clear is

(C)
Welcome to IITM BS DEGREE
First level to clear is Foundation

(D)
Welcome to DEGREE
First level to clear is {{ Level }}

Official Answer Key: 6406533361677

MAD1_281 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

Consider the following Python code and select correct output.

```
from jinja2 import Template

class Loan:

   def __ __(self, accno, loan_amount, interest_rate):
       self.accno = accno
       self.loan_amount = loan_amount
       self.interest_rate = interest_rate

   def _ _details(self):
       return (self.accno, self.loan_amount, self.interest_rate)

loan1 = Loan(101, 45900, 5)
loan2 = Loan(102, 170000, 6)

tm1 = Template("Account1 : {{l1.get_loan_details()}}")
tm2 = Template("Account2 : {{l2.get_loan_details()}}")

print(tm1.render(l1=loan1))
print(tm2.render(l2=loan2))
```

Options :

(A) **Account1 : (102, 170000, 6)**
Account2 : (101, 45900, 5)

(B) **Account2 : (102, 170000, 6)**

(C) **Account1 : (101, 45900, 5)**

(D) **Account1 : (101, 45900, 5)**
Account2 : (102, 170000, 6)

Official Answer Key: 6406533361716

MAD1_291 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

Consider the following code block and select the correct option regarding the output when we run the python file app.py.

app.py

```
from jinja2 import Template

template=  '
           e to {Degree} DEGREE
           level to clear is {{level}}
           '

rendered=Template(template)
output=rendered.render(Degree="IITM BS",level="Foundation")
print(output)
```

Options :

(A)
Welcome to IITM BS DEGREE
First level to clear is {{ Level }}

(B)
Welcome to {Degree} DEGREE
First level to clear is

(C)
Welcome to IITM BS DEGREE
First level to clear is Foundation

(D)
Welcome to DEGREE
First level to clear is {{ Level }}

Official Answer Key: 6406533362461

MAD1_294 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

Consider the following Python code and select correct output.

```
from jinja2 import Template

class Loan:

   def __init__(self, accno, loan_amount, interest_rate):
       self.accno = accno
       self.loan_amount = loan_amount
       self.interest_rate = interest_rate

   def get_loan_details(self):
       return (self.accno, self.loan_amount, self.interest_rate)

loan1 = Loan(101, 45900, 5)
loan2 = Loan(102, 170000, 6)

tm1 = Template("Account1 : {{l1.get_loan_details()}}")
tm2 = Template("Account2 : {{l2.get_loan_details()}}")

print(tm1.render(l1=loan1))
print(tm2.render(l2=loan2))
```

Options :

(A) Account1 : (102, 170000, 6)
Account2 : (101, 45900, 5)

(B) Account2 : (102, 170000, 6)

(C) Account1 : (101, 45900, 5)

(D) Account1 : (101, 45900, 5)
Account2 : (102, 170000, 6)

Official Answer Key: 6406533362500

• 4.2 Jinja2 Filters, Macros & Template Inheritance (12 Questions)

MAD1_005 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

What will be the output of the following Jinja2 template when rendered in Flask?

```
{% macro greet(name) %}
Hello, {{ name }}!
{% endmacro %}

<!DOCTYPE html>
<html>
<head>
    <title>Jinja2 Macro Test</title>
</head>
<body>
    {{ greet("Alice") }}
    {{ greet("Bob") }}
</body>
</html>
```

Options :

(A) Hello, Alice! Hello, Bob!

(B)
```
jinja2.exceptions.UndefinedError: 'greet' is undefined
```

(C) Hello, Alice!

Hello, Bob!

(D) {{ greet("Alice") }}

{{ greet("Bob") }}

Official Answer Key: 6406534366195

MAD1_098 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

What will be the output of the following Jinja2 template when rendered in Flask?

```
{% macro greet(name) %}
Hello, {{ name }}!
{% endmacro %}

<!DOCTYPE html>
<html>
<head>
    <title>Jinja2 Macro Test</title>
</head>
<body>
    {{ greet("Alice") }}
    {{ greet("Bob") }}
</body>
</html>
```

Options :

(A) Hello, Alice! Hello, Bob!

(B)
```
jinja2.exceptions.UndefinedError: 'greet' is undefined
```

(C) Hello, Alice!

Hello, Bob!

(D) {{ greet("Alice") }}

{{ greet("Bob") }}

Official Answer Key: 6406534364653

MAD1_113 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

What will be the output of the following Jinja2 template when rendered in Flask?

```
{% macro greet(name) %}
Hello, {{ name }}!
{% endmacro %}

<!DOCTYPE html>
<html>
<head>
    <title>Jinja2 Macro Test</title>
</head>
<body>
    {{ greet("Alice") }}
    {{ greet("Bob") }}
</body>
</html>
```

Options :

(A) Hello, Alice! Hello, Bob!

(B)
```
jinja2.exceptions.UndefinedError: 'greet' is undefined
```

(C) Hello, Alice!

Hello, Bob!

(D) {{ greet("Alice") }}

{{ greet("Bob") }}

Official Answer Key: 6406534365424

MAD1_146 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535115574

MAD1_159 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535116417

MAD1_170 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535118348

MAD1_182 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(C) 1 → B, 2 → C, 3 → D, 4 → A

(D) 1 → A, 2 → B, 3 → C, 4 → D

(E) 1 → B, 2 → D, 3 → C, 4 → A

(F) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535130968

MAD1_194 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(C) 1 → B, 2 → C, 3 → D, 4 → A

(D) 1 → A, 2 → B, 3 → C, 4 → D

(E) 1 → B, 2 → D, 3 → C, 4 → A

(F) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535131864

MAD1_205 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(C) 1 → B, 2 → C, 3 → D, 4 → A

(D) 1 → A, 2 → B, 3 → C, 4 → D

(E) 1 → B, 2 → D, 3 → C, 4 → A

(F) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535132760

MAD1_216 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535125841

MAD1_228 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

Official Answer Key: 6406535128333

MAD1_239 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

Options :

(B) 1 → B, 2 → C, 3 → D, 4 → A

(C) 1 → A, 2 → B, 3 → C, 4 → D

(D) 1 → B, 2 → D, 3 → C, 4 → A

(E) 1 → C, 2 → D, 3 → A, 4 → B

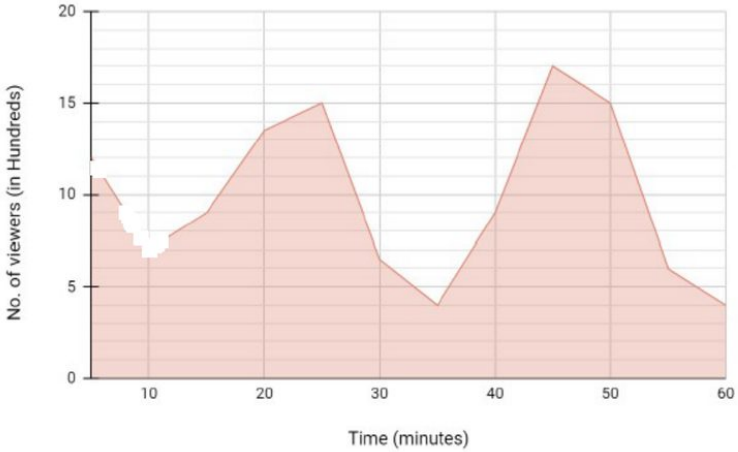
Official Answer Key: 6406535129903

• 4.4 Database Integration in Flask (sqlite3, Cursor & Connection) (1 Questions)

MAD1_251 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

A certain video that was available on the web only for an hour occupied 2 gigabytes of memory of the server. If the number of viewers of the video over time for that hour is given by the curve shown below and assuming that each viewer required an individual connection to the server to view the video, what would be the minimum RAM requirement of the server that could process requests from all the viewers simultaneously at any point of time?



Options :

(E) 1.7 GB

Official Answer Key: 6406532239862

• 4.5 Full-Stack CRUD Workflows & Form Processing (1 Questions)

MAD1_127 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

What will be the output of the following Python code?

```
from jinja2 import Template
data = {     : 101, "prod_name": "Groceries", "sales_amt": 5600,
          : 5}

def     (amount, p):
   return amount * p / 100

a = foo(data[  ales_amt"], data["sales_bonus"])
t = Template(
                ns, you got bonus {{ bonus }} by selling product
                me'}}"
)
output = t.render(data=data, bonus=a)
prir`(output)
```

Options :

(E) `Congratulations, you got bonus 70.0 by selling product Groceries`

(F) `Congratulations, you got bonus 140.0 by selling product Groceries`

(7) `Congratulations, you got bonus 280.0 by selling product Groceries`

(8) `Congratulations, you got bonus 560.0 by selling product Groceries`

Question Number : 37 Question Id : 640653825034 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Consider the below table.

Column 1	Column 2
i) Primary Key	a) Method of the request
ii) Foreign key	b) Provides additional functionality at top of original function
iii) Nullable	c) Refers to another parent table column
iv) POST	d) Unique identification of row
v) @<decorator>	e) Undefined value

Use the above table. Which of the below match is correct?

Options :

(A) - d, ii - b, iii - a, iv - e, v - c

(B) - d, ii - c, iii - e, iv - a, v - b

(C) - c, ii - d, iv - e, iii - a, v - b

(D) - c, ii - d, iv - e, iii - b, v - a

Sub-Section Number : 3

(E) 640653122721

Question Shuffling Allowed : Yes

Official Answer Key: 6406532774784